CATS User Manual

Contents

1 Revision History

Date	Version	Comment
15 Dec 2022	1.0.0	Initial release
05 Mar 2023	1.1.0	Addressing first round of feedback; added telemetry explanations
16 Jul 2023	2.0.0	Added explanations for new hardware version; added support for sensor tab on the ground station; added explanations for test button; added bootloader explanations on ground station
21 Jan 2026	2.1.0	Added regulatory info; Added more information on software updates and flight recorder settings; fixed typos
21 Jun 2026	2.1.1	Added section on Ground Station USB Data Streaming
09 Aug 2026	2.1.2	Corrected Ground Station flash memory capacity and CATS Vega dimensions

Table 1: Revision History

Glossary

Apogee The highest point of the rocket during a parabolic flight. 9, 12, 18, 19, 24, 39, 41

Barometer A sensor that measures ambient barometric pressure. 10, 31, 47

Calibrating A flight state in which the flight computer is in safe mode and performs no actions. 12, 30, 49

CLI Command Line Interface, used to read and change flight-computer values without the Configurator. 16, 26, 53

CRC Cyclic Redundancy Check, used to verify that received data is not corrupted. 48

DFU Device Firmware Update, a mode used to flash new firmware to an embedded system. 38

Drogue Chute A relatively small parachute deployed at apogee to stabilize and slow the rocket during descent. 24

FHSS Frequency Hopping Spread Spectrum, a protocol that transmits telemetry data across multiple carrier frequencies. 48

FreeRTOS An operating system for embedded CPUs and the backbone of the CATS software. 47

FSM Finite State Machine, a model that describes the flight phases, how transitions between them occur, and which transitions are permitted. 12

GNSS Global Navigation Satellite System, such as GPS or Galileo. 11, 21, 34

I/O Input/Output, a collective term for the system's inputs and outputs. 9, 11, 21, 22, 23, 41

IMU Inertial Measurement Unit, a sensor that measures linear acceleration and angular velocity. 10, 12, 47

Kalman Filter A filtering technique used to estimate states according to physical laws and measurements. 49

Liftoff The event that marks the start of powered ascent. 12, 13, 16, 19, 21, 24, 49, 50

Main Chute The larger parachute deployed at a specified height above ground level to slow the rocket before landing. 16, 24

Patch Antenna A flat, highly directional antenna used by the CATS Vega to receive GNSS signals. 21

Power Supply A source of electrical power, typically a battery in a rocket. 9, 22

PWM Pulse Width Modulation, a method used to control most modern servos. 22

Pyro A pyrotechnic charge initiated by applying current through two electrical leads. 10, 11, 12, 13, 18, 21, 22, 23, 39, 41

Ready A flight state; when in this state, liftoff can be detected. 12, 21, 25, 30, 49

RF Radio Frequency. 21

Servo A small electric actuator whose position is controlled by a PWM signal. 10, 12, 13, 16, 21, 22, 23, 24, 41

Touchdown The final flight phase, entered when the rocket lands. 12

UART A serial communication interface used to exchange data between devices. 10, 11

2 Disclaimer

The use of the CATS System is at your own risk. The CATS System must always be used in conjunction with a second safety system that operates in a different manner, such as motor ejection or another electronic system, to ensure maximum safety.

The manufacturer is not liable for any damages that may occur as a result of using the CATS System, and will not be held responsible for any damages inflicted on third parties.

Do not touch the CATS Vega while it is powered through the battery port because high currents flow through the board.

The manufacturer cannot be held liable for any program errors or malfunctions in the software.

The hardware warranty covers manufacturing errors and defects in workmanship and materials for a period of two years from the date of purchase. This warranty does not cover damage caused by improper handling, accidents, or other external factors. In the event of a covered manufacturing error or defect, the manufacturer will repair or replace the hardware at no cost to the user. To make a claim under this warranty, the user must provide proof of purchase and report the issue to the manufacturer within the warranty period. The manufacturer reserves the right to inspect the hardware and determine the cause of any reported issue before providing a repair or replacement. This warranty is non-transferable and only applies to the original purchaser of the hardware.

3 Important Regulatory Information

3.1 FCC Part 15 Compliance

This equipment has been designed to operate in accordance with Part 15 of the FCC Rules for unlicensed radio frequency devices. To maintain cooperative use of the shared spectrum, operation should be subject to two general conditions: (1) the device should not cause harmful interference, and (2) it must accept any interference received, including interference that may cause undesired operation.

3.2 Intended Use and Operational Constraints

The CATS Vega and Ground Station are developed primarily for educational, professional, and experimental use within amateur High Power Rocketry. In the United States, rockets of this class are considered aircraft by the Federal Aviation Administration (CFR 14 §101.25). Operators should adhere to FAA and National Fire Protection Association (NFPA) safety codes, which typically require maintaining a safe distance of at least 1,500 feet from populated buildings or uninvolved persons. Due to the high-power radio frequency emissions in the 2.4 GHz band, users are strongly advised against operating this telemetry system in residential areas to prevent potential disruptions to local consumer networks.

3.3 Hardware Certification and Radio Frequency (RF) Operations

The system utilizes Semtech SX1280 RF modules operating in the 2.4 GHz Industrial, Scientific, and Medical (ISM) band. As this operates on a shared, unlicensed band, users are not protected against interference from other ambient RF sources. We have made good faith efforts to ensure the hardware design follows standard EMC practices; however, should your system cause interference with licensed radio systems or residential Wi-Fi networks, you are expected to power down the device until the interference issue is resolved.

3.4 European Union (EU) and CE Compliance Notice

For operation within the European Union (EU) and the European Economic Area (EEA), this equipment is subject to the Radio Equipment Directive (RED) 2014/53/EU. Users operating within the EU must configure the telemetry output via the CATS Configurator to comply with the ETSI EN 300 328 harmonized standard, which strictly limits 2.4 GHz wideband transmissions to a maximum of 100 mW (20 dBm) Equivalent Isotropic Radiated Power (EIRP).

Exception for Licensed Amateur Radio Operators: Users holding a valid amateur radio license are authorized to operate at higher power levels within the 2.4 GHz amateur allocation. Such operation must strictly adhere to national amateur radio regulations, including station identification rules and prohibitions on encrypted commercial data.

3.5 Environmental and Safety Declarations

This product is factory-assembled utilizing lead-free manufacturing processes with the intent of aligning with global Restriction on Hazardous Substances (RoHS) principles. Please practice responsible disposal at the end of the product's life cycle by recycling it as electronic waste rather than disposing of it in standard household trash.

4 Introduction

Welcome to the CATS flight computer! The following pages explain the CATS System, which consists of the CATS Vega flight computer and the CATS Ground Station, so that you can use it in your rocket. Each component has a How to Use section that provides the information needed to launch your rocket. Other sections provide advanced information that is not required for basic use of the CATS System.
If you have any feedback, suggestions, or need further help with the CATS System, do not hesitate to contact us on our Discord server[1]. This is the first release of the manual and will be improved based on your feedback!
The entire CATS ecosystem is open source. You can find all code and hardware designs on our GitHub page[2].

4.1 Coverage of This Manual

This manual covers the use of the CATS Vega flight computer and its Ground Station. It explains how the flight computer works, how to connect it to a Power Supply and deployment actuators, and how to configure it for your flight trajectory.
This manual does not cover everything that can be done with the CATS System. In particular, it does not explain how to modify the software or hardware, or how the software works in detail. For further information about those topics, contact us on our Discord server 1 .

4.2 Module Overview

The CATS System has three main components.

CATS Vega: The flight computer installed inside your rocket.
Ground Station: The receiver that displays flight data in real time and helps you track your rocket after it has landed.
Configurator: The computer application used to configure the CATS Vega and visualize completed flights.

4.3 Basic Functionality

This section provides a broad overview of the CATS System for users who are new to rocketry. The following sections explain each topic in greater detail.
The CATS Vega is the core of the system. Its main purpose is to deploy the drogue chute at Apogee and the main chute at a user-configurable height above ground level. To do this, it combines barometric and accelerometer data to estimate height above ground level and velocity.
The CATS System defines five **events**: *Liftoff*, *Burnout*, *Apogee*, *Main Deployment*, and *Touchdown*. The onboard control system triggers these events automatically, while the user defines the actions performed when each event occurs.
Up to eight **actions** can be assigned to each event. Actions include delays, pyro-channel and servo-channel triggers, and low-level I/O signals. This allows the user to define how the deployment mechanism is actuated. Most commercial recovery mechanisms use pyro or servo channels, and the CATS Vega provides two of each.
At the same time, the system transmits important data to the Ground Station, including GNSS data, current height, velocity, and flight state. The data is also recorded in onboard flash memory.
The Ground Station is also used for testing mode, in which the user can trigger configured events to test separation mechanisms.

[1] https://discord.gg/r7ErmSNvsy
[2] https://github.com/catsystems

5 CATS Vega

This section describes how the flight computer works and how to configure it for a flight. The How to Use section explains its basic features. For more detailed information, refer to Section 9 .

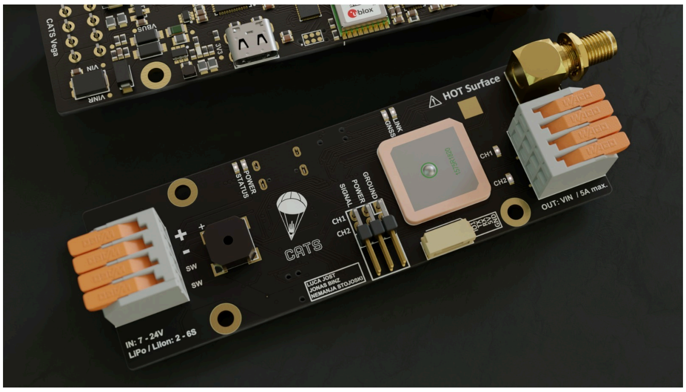


Figure 1: CATS Vega

5.1 Hardware

5.1.1 Specifications

Size	100 × 33 × 15 mm (without the antenna)
Weight	33 g
Input Voltage	7 - 24 V
Power Consumption	100 mA
Number of Pyro channels	2
Number of Servo channels	2
Number of IOs	1
Additional IO	UART
Servo Power	5 V / 3 A max.
Microcontroller	STM32F4
Flash Memory	16MB
IMU	LSM6DSO32
Barometer	MS5607
Radio Frequency	ISM 2.4GHz
Radio Power	Up to 1W
Radio Range	Tested to 10km @100mW

Table 2: Vega Specifications

5.1.2 Hardware Overview

This section provides a quick hardware overview and shows the location of each port. The numbered markers in Figure 2 correspond to the following list.

1. **Switch Port**; Connect a manual switch between the two terminals.

2. **Battery Port**; Connect a battery to these terminals and observe the correct polarity.

3. **Buzzer**; Indicates flight-computer readiness and status through beeping patterns, as explained in Section 5.4 .

4. **Status LEDs**; The POWER LED is illuminated when power is present. The STATUS LED blinks when the system is operating normally.

5. **USB Connector**; The connector is on the other side of the board.

6. **Test Button**; If this button is held during startup, the board enters testing mode if a testing phrase has been configured.

7. **Servo Connector**; This connector fits the standard servo connectors. Two servos can be connected to this connector.

8. **Telemetry LEDs**; The GNSS LED blinks whenever GNSS coordinates are received. The LINK LED blinks after a connection to the Ground Station has been established.

9. **Low-Level I/O and UART Connector**; Connect external hardware to this port to exchange data with the CATS board.

10. **Pyro LEDs**; These red LEDs are turned on when continuity of the Pyro channel is detected.

11. **Pyro Channel 1**; Connect a pyrotechnic charge or another supported device to this connector.

12. **Pyro Channel 2**; Connect a pyrotechnic charge or another supported device to this connector.

13. **Antenna Connector**; Connect an antenna here so that the CATS Vega can transmit data to the Ground Station.

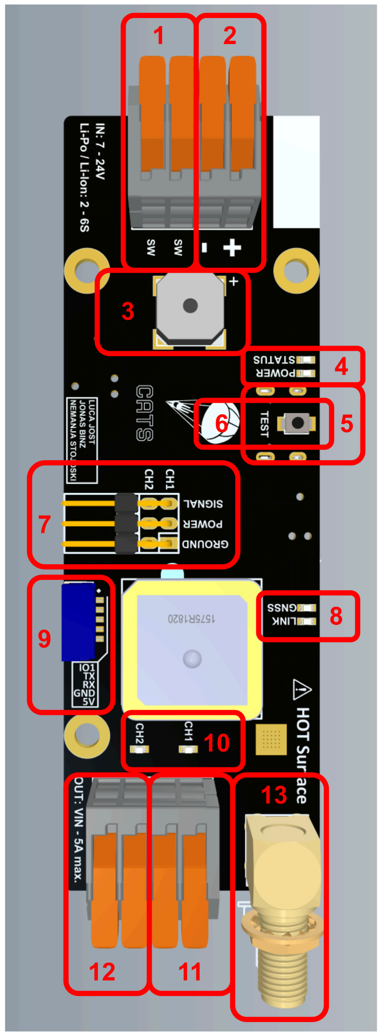


Figure 2: CATS Vega board hardware specifications

5.2 Working Principle

This section briefly introduces the operating principles needed to understand the Vega flight computer's configuration options.

5.2.1 Configurable Actions & Finite State Machine

The finite state machine (FSM), shown in Figure 3 , controls the outputs of the Vega flight computer. When the flight computer is turned on, it starts in the Calibrating state. Every flight follows the sequence of states shown in Figure 3 . Whenever a state transition occurs, the associated event is triggered.

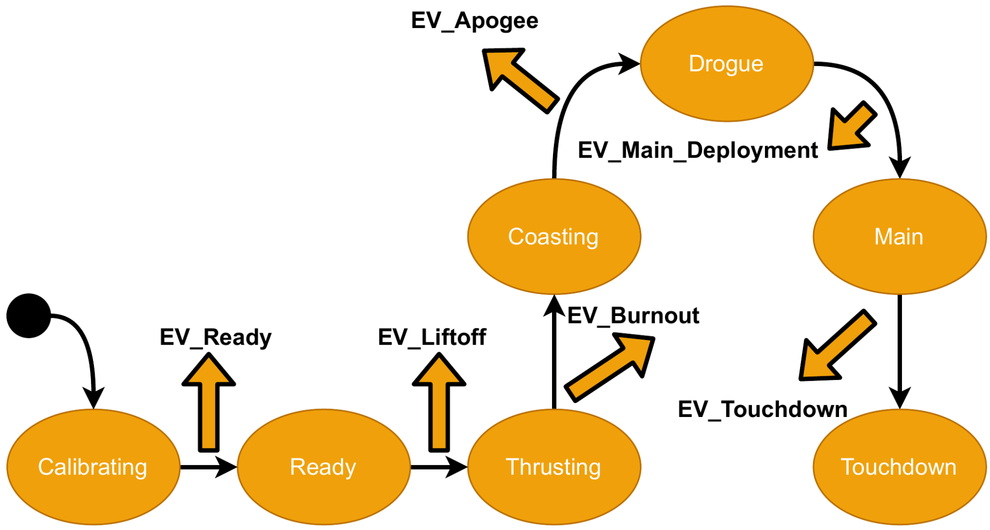


Figure 3: Finite State Machine controlling the CATS software.

Each event can trigger one or more actions, such as a Pyro channel, Servo channel, or timer. Use the Configurator to assign actions to events, as described in Section 5.3.2 .

Calibrating → Testing	A telemetry command starts testing mode. This transition is available only if testing mode was enabled through the Configurator.
Calibrating → Ready	IMU (gyroscope and linear acceleration) readings are constant for 10 seconds.
Ready → Thrusting	The measured acceleration in any direction exceeds the user-defined acceleration threshold for 0.1 seconds.
Thrusting → Coasting	The measured acceleration in the “up” direction is smaller than 0 $\frac{m}{s^2}$ for 0.1 seconds
Coasting → Drogue	The estimated velocity needs to be smaller than 0 $\frac{m}{s}$ for 0.3 seconds
Drogue → Main	The estimated height is below the user-defined height for 0.3 seconds
Main → Touchdown	The estimated velocity is in the bound $[-3, 3]\frac{m}{s}$ for 1 second

Table 3: FSM Transition Specifications

With this setup for state changes, the flight has a strictly controlled order. The Main event can only be thrown after the Apogee event. Events are also **unique**; during a flight only one event can be thrown.

> **Note:** A software safeguard checks the time between Liftoff and Apogee. If that time is smaller than 1.5 seconds, the flight computer assumes a faulty Liftoff detection and no further events or actions are activated. The flight computer then jumps instantly to Touchdown without triggering anything.

5.2.2 Actions

When an event is triggered, the flight computer performs the actions assigned to it. Up to eight actions can be assigned to each event, supporting a wide range of applications. Examples include:

- Enabling a camera at Liftoff using a Pyro channel,
- Actuating a solenoid valve for two seconds using a Pyro channel,
- Enabling some mechanism at engine burnout,
- Disabling the camera at touchdown using the Pyro channel,
- ...

The full range of actions can be found below.

Action	Parameter
Pyro 1	ON/OFF
Pyro 2	ON/OFF
Servo 1	[0-1000]‰
Servo 2	[0-1000]‰
Low-Level I/O	ON/OFF
Delay	[0-15000] ms
Recorder	ON/OFF/PREFILLING

Table 4: Exhaustive List of all possible Actions

> **Note:** The *PREFILLING* recorder option continuously accumulates log elements in a buffer. Once the recorder transitions into *ON* mode, the elements in the buffer are written to the flash chip. This information can be used to analyze the initial ignition sequence and thrust build-up.

How actions can be configured is shown in section 7 .

5.3 How to Use

Now that the hardware and software have been introduced, this section explains how to configure and mount the flight computer, update its software, and generate plots from flight data.

5.3.1 Connection to Your Computer

Before connecting your CATS Vega to your computer, download the Configurator from our releases page. Drivers are usually not required. If your computer does not recognize the device, refer to the troubleshooting steps in our wiki[3].

5.3.2 Description of the Configurator

The Configurator allows you to configure the CATS System from your computer. Download the latest stable version from the releases page[4]. Release candidates are marked accordingly and should be used only to test new features.

Home Tab

The Home tab appears when the Configurator starts. If the flight computer is already connected to your computer, select the correct communication port (label 2 in Figure 4) and select Connect. If the connection times out, confirm that you selected the correct device. If several flight computers are connected, make sure that you are configuring the intended one. Until a flight computer is connected, the other tabs remain disabled (label 6), and the Configurator indicates that no board is connected (labels 5 and 7).
If the required communication port does not appear in the dropdown menu, select Refresh (label 1). If the port still does not appear, refer to our wiki for troubleshooting steps.

1: Refresh Button	Refreshes the communication ports shown under label 2
2: Com Port Selection	Selects the communication port used to connect to the CATS Vega
3: Connect Button	Connects to the CATS Vega
4: Home Screen	Displays general information about the Configurator
5: Connection Status	Indicates that the flight computer is not currently connected
6: Navigation	Remains disabled until a flight computer is connected, then provides navigation between tabs
7: Connection Status	Shows the connection status
8: Flight Log Graph	Loads a CATS flight log (.cfg) file for plotting. See Section 5.3.7 for more information

Table 5: Overview of the different Settings for the Home Tab

[3]https://github.com/catsystems/cats-embedded/wiki/Installation
[4]https://github.com/catsystems/cats-configurator/releases

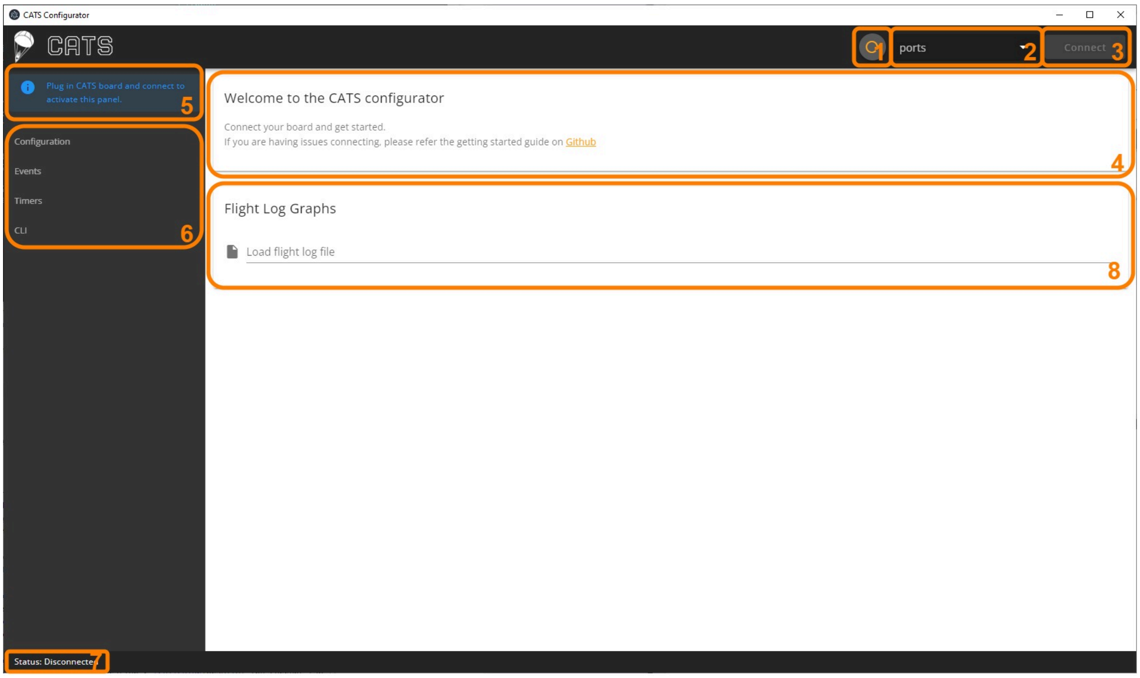


Figure 4: Home Menu.

Configuration Tab

This tab displays the flight computer's status and allows you to configure its parameters. Figure 5 shows the Configuration tab, and the following table explains each label.

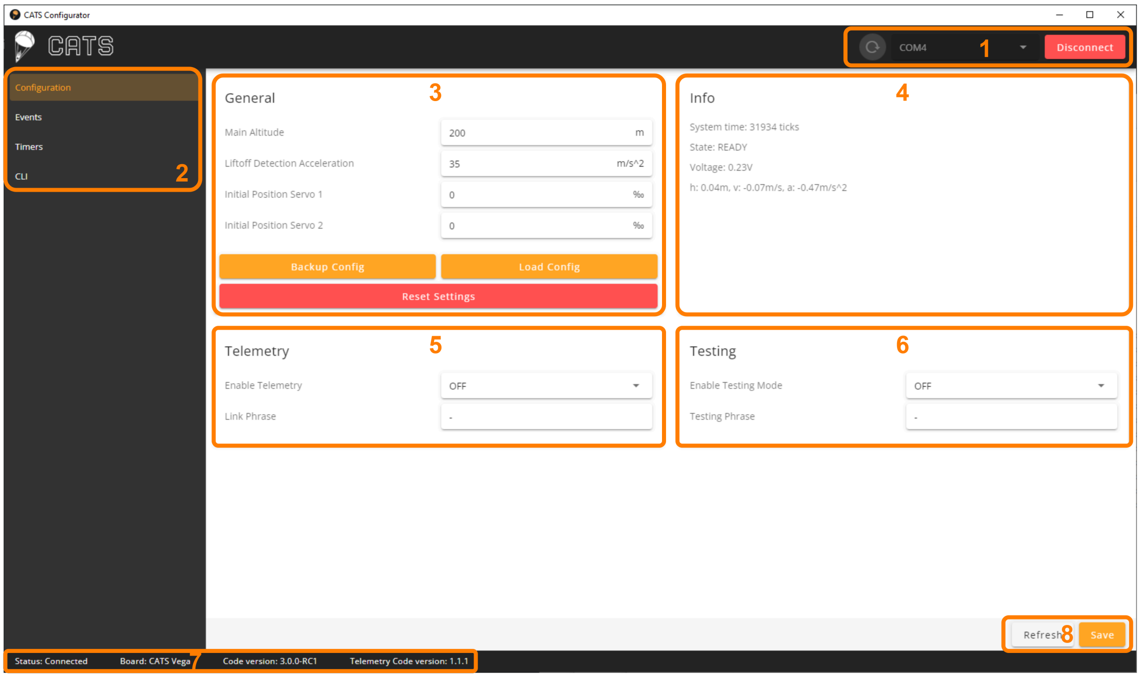


Figure 5: Configuration Menu.

1: Communication Port	Refresh Button	Used to refresh the communication port list
	Com Port	Shows the communication port selected
	Connection Button	Shows the connection status and allows you to connect or disconnect
2: Navigation	Configuration	The current tab
	Events	Event tab, described on the next page
	Timers	Timer tab, described on the next page
	CLI	Access to the CLI for advanced commands, described on the next page
3: General	Main Altitude	Sets the desired height above ground level at which the Main Chute is deployed
	Liftoff Detection Acceleration	User-defined acceleration threshold above which liftoff is detected. We recommend 40 $\frac{m}{s^2}$ for most flights
	Initial Position Servo 1	Initial angle of the Servo connected to Servo port 1. If no Servo is connected, no value is required. At startup, the flight computer drives the Servo to this position.
	Initial Position Servo 2	Same as for Servo 1 but for the second channel.
	Backup Config	Saves a configuration file to your computer. Select Load Config to load this file onto the flight computer.
	Load Config	Loads a previously backed-up configuration from your computer onto the flight computer.
	Reset Settings	Set the default parameters of the flight computer.

Table 6: Overview of the Configuration Tab

4: Info	System Time	The elapsed system time since startup, in milliseconds
	State	The current flight state of the board
	Voltage	The battery voltage (if no battery is connected, a value below 1 V is shown)
	State Estimation	The currently estimated height above ground level, velocity and acceleration
5: Telemetry	Link Phrase	A link phrase containing 4 to 16 characters. It must match the link phrase configured on the Ground Station.
	Enable Telemetry	Set to ON or OFF to enable or disable telemetry.
6: Testing	Enable Testing Mode	If this is set to ON, the flight computer enters testing mode after its next reboot. It must then be armed through telemetry using the Ground Station. Read Section 8 thoroughly before using testing mode.
	Testing Phrase	A testing phrase containing 4 to 16 characters. It must match the testing phrase configured on the Ground Station. Read Section 8 thoroughly before using testing mode.
7: Hardware Info	Status	Connection status of the board
	Board	CATS Board Name
	Code Version	Current code version
	Telemetry Code Version	Current code version on the telemetry chip
8: Save Settings	Save	Save the current settings to the board. **Attention:** if Save is not pressed, the values are not saved to the board!
	Refresh	Refresh the displayed values. When this button is pressed, the saved values from the flight computer are fetched and shown.

Table 6: Overview of the Configuration Tab (Cont.)

Event Tab

This tab shows the actions configured for each event. Section 5.2.1 describes the events in detail. Up to eight actions can be assigned to each event: Liftoff, Burnout, Apogee, Main Deployment, Touchdown, Custom 1, and Custom 2. Select Add Action under an event to assign a new action. To avoid missing flight data, we recommend assigning Recorder On to Liftoff and Recorder Off to Touchdown.
In the example shown in Figure 6 , Liftoff, Apogee, Main Deployment, and Touchdown have mapped actions. At Liftoff, the recorder is enabled. At Apogee, Pyro channel 1 is activated. At Main Deployment, Pyro channel 2 is activated. At Touchdown, the recorder is disabled.
Figure 7 shows the pop-up menu opened by selecting Add Action. Select the action type in the upper section, then configure its behavior in the lower section.
To remove an action, select the cross beside it. Select the gear icon to reconfigure an action. Select **Save** after making changes.

> **Note:** The custom events can only be triggered as described below in the Timers section.

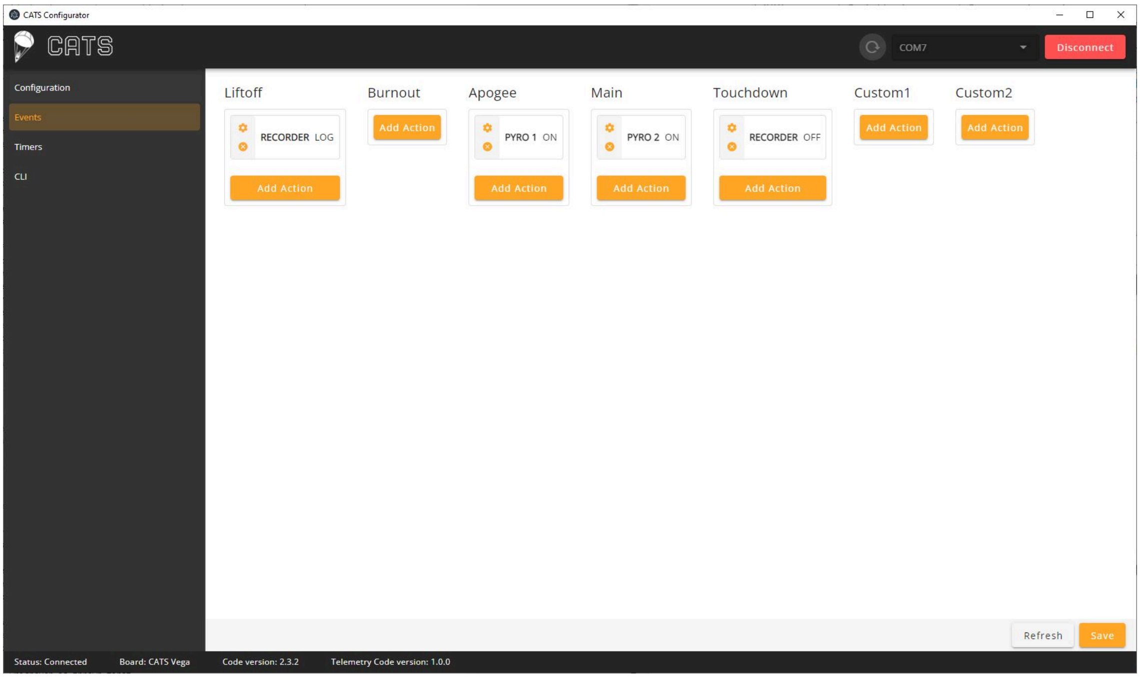


Figure 6: Event Menu.

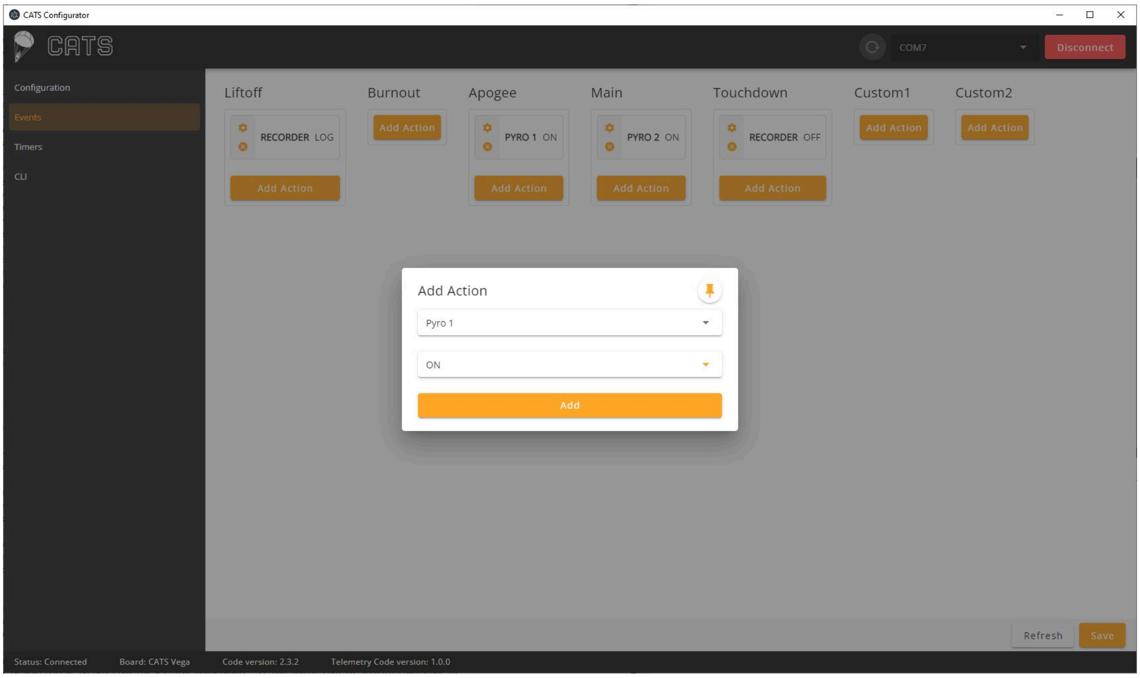


Figure 7: Configuring an Event in the Event Menu.

Timers

Use this tab to configure the four available timers. Enable or disable each timer with the yellow button in the upper-right corner of its panel.
After enabling a timer (Timer 1 in Figure 8), configure its start event, duration in milliseconds, and end event. In the example, the start event is Liftoff, the duration is 10000 ms, and the end event is Apogee. At liftoff, a 10-second timer starts and triggers the Apogee event when it expires. This triggers the event without placing the flight computer in the Apogee state.

Note: Events are unique, meaning that if a timer is used to trigger apogee, only one apogee event will be thrown by the flight computer. It will either be the timer or the event from the estimation, whichever is thrown first.

Custom Events

Two custom events are available. To activate one, configure a timer whose end event is the desired custom event. This allows you to execute actions at an arbitrary time.
This is particularly useful for payload experiments. For example, to actuate a device 10 seconds after apogee, configure a timer that starts at apogee, runs for 10 seconds, and triggers Custom 1. Then assign the desired actions to that event.

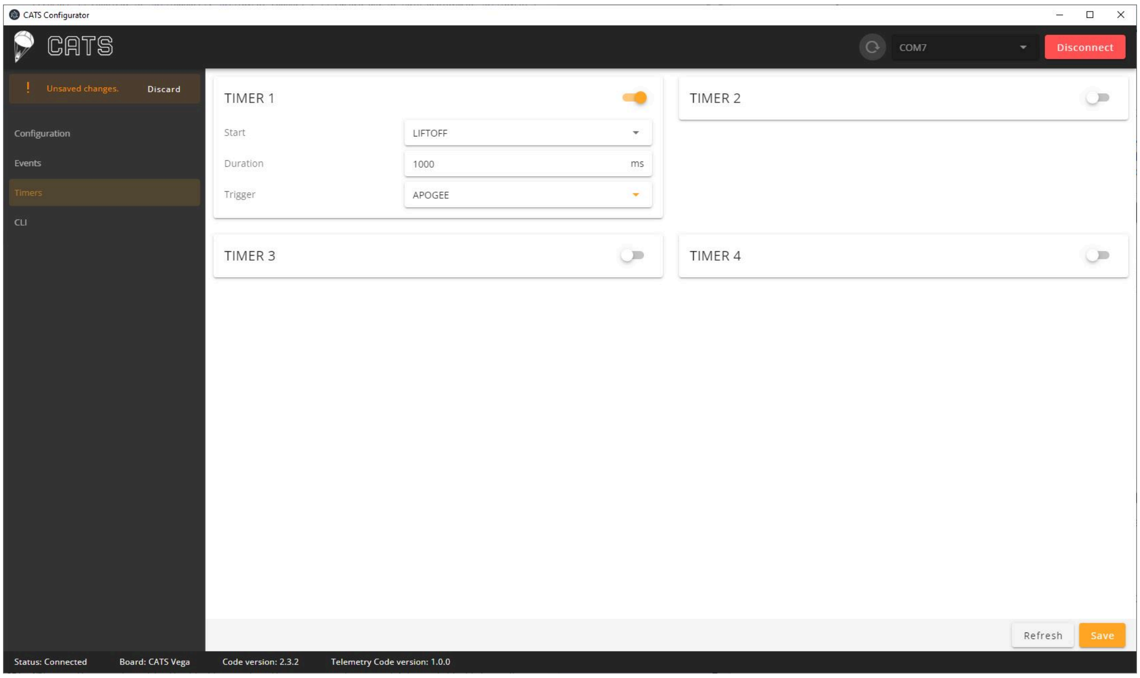


Figure 8: Timer Menu.

CLI

The CLI tab allows the user to send commands directly to the CATS board. Section 9.5 explains all supported commands. Figure 9 shows the CLI.

Figure 9: Command Line Interface.

5.3.3 Mounting

The CATS Vega **does not** require a specific mounting orientation. The system automatically detects the gravity vector for internal state estimation, so you can mount the board in any orientation.
The board has a length of 100 mm, a width of 33 mm and a total height of 15 mm. Three mounting holes secure the system to the rocket. The mounting holes are spaced 60 mm by 27 mm and are designed for M3 screws. Use spacers to prevent the electronics from touching the rocket. Download the system's 3D files from our GitHub repository[5].
For reliable radio reception during flight, pay close attention to the area surrounding each antenna. Install the CATS flight computer in a radio-transparent section of the rocket, such as fiberglass or cardboard. Do not install it in a carbon-fiber section, which blocks RF signals. Ensure that the onboard Patch Antenna has a clear view of the sky for optimal GNSS reception, and keep the telemetry antenna away from metal objects.

After power-up, the system detects the up direction once it is stable. A beeping pattern and the Ground Station indicate when the flight computer enters the Ready state. In this state, the flight computer is armed and waiting for Liftoff. Do not move the rocket, and follow all safety guidelines. At this stage, the flight computer can be disarmed only by switching it off. For more information about calibration, refer to Sections 9.3 and 5.2.1 .

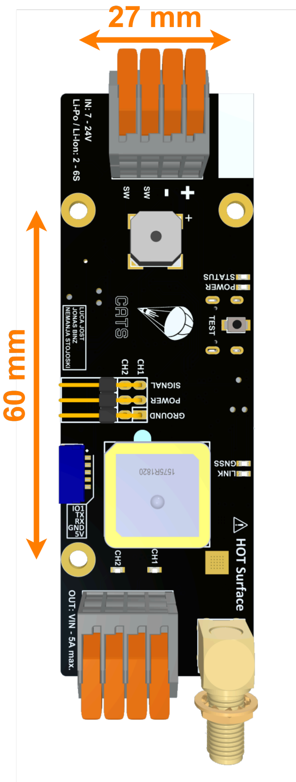


Figure 10: CATS Vega board with the mounting hole and dimensions.

Warning: Power up the flight computer only after the rocket is upright on the launch pad. Calibration is performed once, as soon as no motion is detected after startup.

5.3.4 Battery, Switch and Actuators

The CATS Vega has one battery port, one switch port, two Pyro channels, two Servo channels, and one low-level I/O. The following sections briefly explain each port. Table 7 summarizes the most important parameters. For more information about the board hardware, refer to Section 5.1 and the labeled board diagram in Figure 2 .

Battery Port

The battery port supports voltages between 7 and 25 volts. For LiPo and Li-ion batteries, this corresponds to 2- to 6-cell battery packs. The battery port is protected against reverse polarity.

Switch Port

The switch port allows the user to add a mechanical switch to the system. If this switch is turned off, the system is fully disconnected from power.

[5] https://github.com/catsystems/cats-hardware/tree/main/CATS-Vega/3D

Note: The battery current is routed through the switch. Make sure that the wires and the switch are rated for the currents required.

Pyro Channels

The Pyro channels apply the battery voltage to the connected circuitry with a voltage drop of approximately 1 V. An electric match is normally connected to a channel to ignite a black-powder charge. The channels can also power other devices. For example, they can actuate solenoid valves (with an external flyback diode), power cameras, or power other electronic circuits. By default, the maximum continuous current is approximately 1 A. The channels are short-circuit protected by a resettable PTC fuse. This current is more than sufficient to ignite electric matches before the fuse reduces it. If the connected load requires more current, the fuse can be bypassed with a solder jumper on the back of the board. In this configuration, stay below 5 A continuous or 20 A burst. Exercise extreme caution: a short circuit on the channel can damage the board.

Warning: The pyro channels are short-circuit protected by a resettable PTC fuse. If the connected circuit requires more than 1 A, close the solder bridge on the back of the board.

Servo Channels

The Servo channels can actuate PWM Servos. An onboard voltage regulator reduces the battery voltage to 5 V to power the servos. The microcontroller's power rail is completely separate from the 5 V Power Supply therefore, a short circuit on the servo power rail does not affect the system. A maximum current of 3 A can be drawn. A PWM signal is always applied to each Servo channel, and the endpoints can be changed in the Configurator.

Low-Level I/O

The low-level I/O can send a signal to another system. The voltage level is 3.3 V, and the pin is connected directly to the microcontroller. Therefore, the I/O should be used **only** for signal transmission, not to actuate a recovery mechanism.

I/O Specification

I/O	Description	Limits
Battery Port	Connect battery	7-24 V
Switch Port	Connect mechanical switch	n.a.
Pyro Channels	Connect up to two pyrotechnic charges or other devices	Battery voltage / 5 A
Servo Channels	Used for Servo actuation, up to two Servos	5 V / 3 A
Low-Level I/O	Use only for signal transmission, not actuation	3.3 V / 10 mA

Table 7: Overview of the I/Os

5.3.5 Setting up the Minimal Flight Configuration

For nominal flight performance, several parameters must be configured before every flight. In particular, the user must know:

- Expected maximum acceleration
- Recovery mechanism for the Drogue Chute
- Recovery mechanism for the Main Chute
- Time until Apogee (optional)
- Desired deployment altitude of the Main Chute
- Time until Main Chute deployment (optional)

With this information, the user can configure the flight computer. Timers are optional and should be used only as a backup.

1. Connect the flight computer to your computer.

2. Open the Configurator and connect to the board as described in Section 5.3.2 .

3. In the Configuration tab, set the Liftoff threshold. We recommend using a Liftoff acceleration threshold of $40\frac{m}{s^2}$, but make sure that it is around $20\frac{m}{s^2}$ lower than your maximum expected acceleration.

4. In the Configuration tab, set the main altitude to your desired height. This is the height above ground level where the Main Chute will be deployed.

5. If you use a Servo channel in either of your recovery mechanisms, it is now also the time to set the initial Servo position.

6. In the Configurator's Configuration tab, set the link phrase for your CATS Vega.

7. Make sure that the Testing Mode is disabled.

8. Save the settings.

9. Go to the Events tab.

10. For the apogee event, set your deployment mechanism as described in 5.3.2 .

11. For the main deployment event, set your deployment mechanism as described in 5.3.2 .

12. Save the settings.

13. (Optional) Go to the Timers tab.

14. (Optional) Set the Timer One start event to Liftoff and the Timer One end event to apogee. Set the time until apogee with 1-2 seconds margin.

15. (Optional) Set the Timer Two start event to Liftoff and the Timer Two end event to main deployment. Set the time until main deployment with 10-60 seconds margin, depending on the flight time.

16. Save the settings.

17. Set the same link phrase on your Ground Station. Navigate to Settings, select Link Phrase, and enter the same phrase.

The flight computer is now ready to be installed in the rocket. For this flight configuration, complete the following steps:

1. Mount the flight computer to your rocket.
2. Connect the switch to the switch port.
3. Connect the battery to the battery port.
4. Connect the recovery mechanism for the apogee event.
5. Connect the recovery mechanism for the main event.
6. Place the rocket on the launch pad.
7. Turn on the flight computer with the switch.
8. The Ground Station will begin receiving data.
9. Wait for the flight computer to finish calibrating and show READY on the Ground Station.
10. The flight computer is now armed. Every 6 seconds, the flight computer beeps twice to indicate that it is in the Ready state.
11. Launch your rocket!

5.3.6 How to Get the Data on Your Computer

After the flight, connect the board to a computer with a USB-C cable. The flight computer appears as a USB drive, allowing you to drag and drop the flight data onto your desktop. You can also drag the flight log directly into the Configurator to plot it.

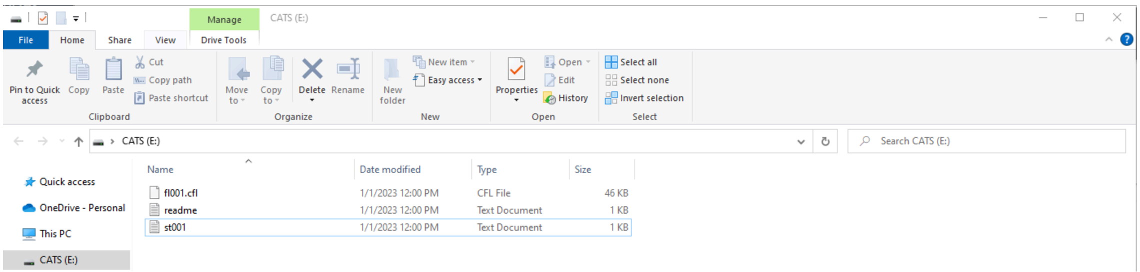


Figure 11: USB drive when the CATS Vega is plugged into the user computer.

5.3.7 Visualizing the Flight Data

To visualize flight data, open the Configurator and drag and drop the CATS flight log (`.cfg`). The Configurator immediately plots the important data: altitude, velocity, acceleration, angular velocity (x, y, and z), linear acceleration (x, y, and z), pressure, state changes, and actions.
You can export the plots as `.html` files, which retain their zoom functionality when opened in a browser, or as `.csv` files for further processing.

A legacy Python plotting tool is also available as a reference for custom implementations. It is described briefly in Section 9.4 .

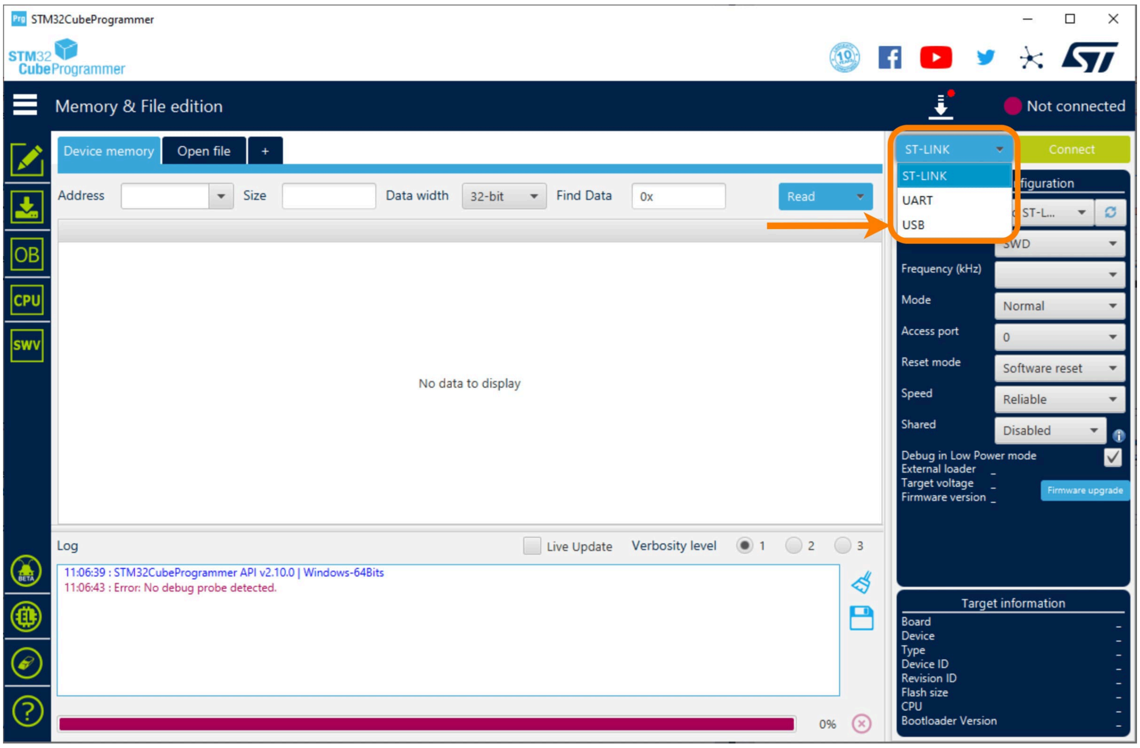
5.3.8 Software Updates

The software is continuously improved, so install each new update when it is released. Updates are announced on our Discord server[6]. To update the software, follow these steps:

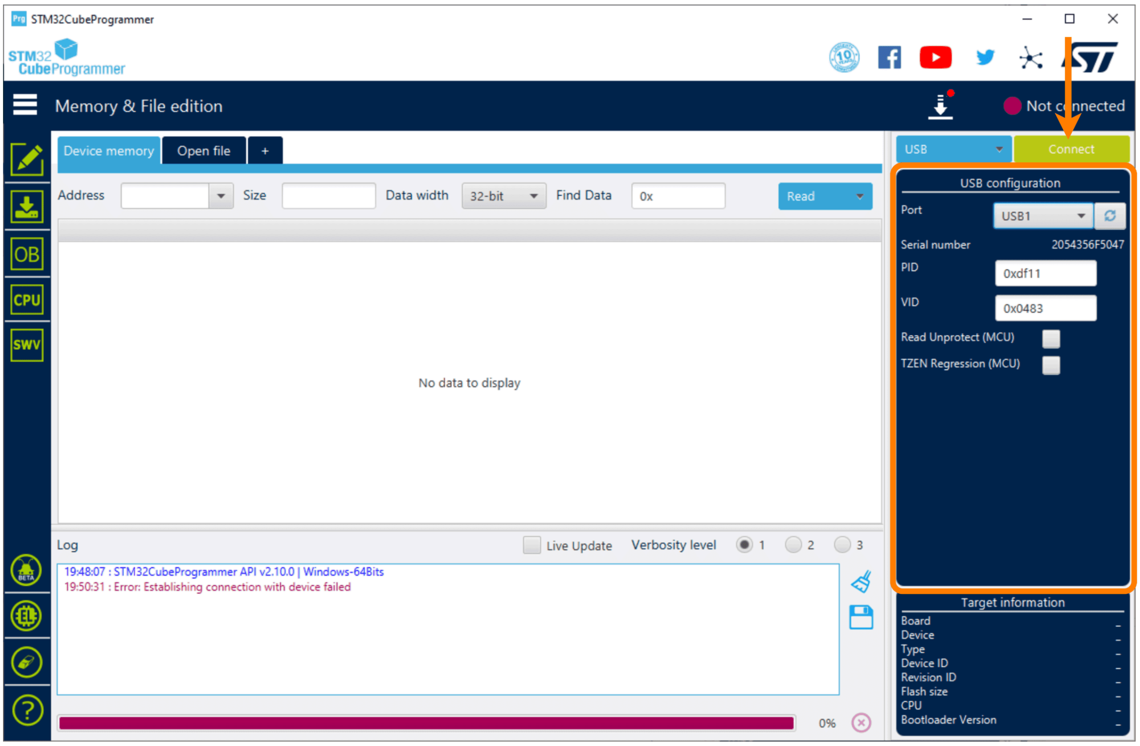
1. Download and install the STM Programmer[7] (STM32CubeProg).
2. Connect the CATS Vega with a USB-C cable.
3. Start the Configurator, select the correct COM port, select Connect, and open the CLI tab described in Section 5.3.2 .
4. In the CLI, enter `bl` and send the command.
5. The CATS Vega will disconnect. Close the Configurator.
6. Start the STM32 Programmer.
7. In the upper-right corner of the programmer, select USB (Figure 12a).
8. On the right, select the appropriate USB port (Figure 12b).
9. Select Connect (Figure 12b).
10. Confirm that the upper-right corner shows that the programmer is connected to the board (Figure 12c).
11. In the left navigation panel, select Erasing & Programming (Figure 12d).
12. In the File path field, select the firmware file to flash (the filename ends in `.bin`). The latest release of the CATS software is available here[8] (Figure 12d).
13. Select Start Programming (Figure 12d).
14. Wait for the "File Download Complete" pop-up (Figure 12e).
15. Disconnect and reconnect the board. Start the Configurator and verify that the version number has been updated.
16. You've successfully updated the software!

Warning: The steps above update only the Vega's flight-control software (`flight_computer.bin`). The telemetry code (`telemetry.bin`) resides on another chip and cannot be updated via USB. Updating the telemetry software requires an STLINK-V3MINI debugger and a TC2030-IDC-NL 6-pin connector.

[6] https://discord.gg/r7ErmSNvsy
[7] https://www.st.com/en/development-tools/stm32cubeprog.html
[8] https://github.com/catsystems/cats-embedded/releases

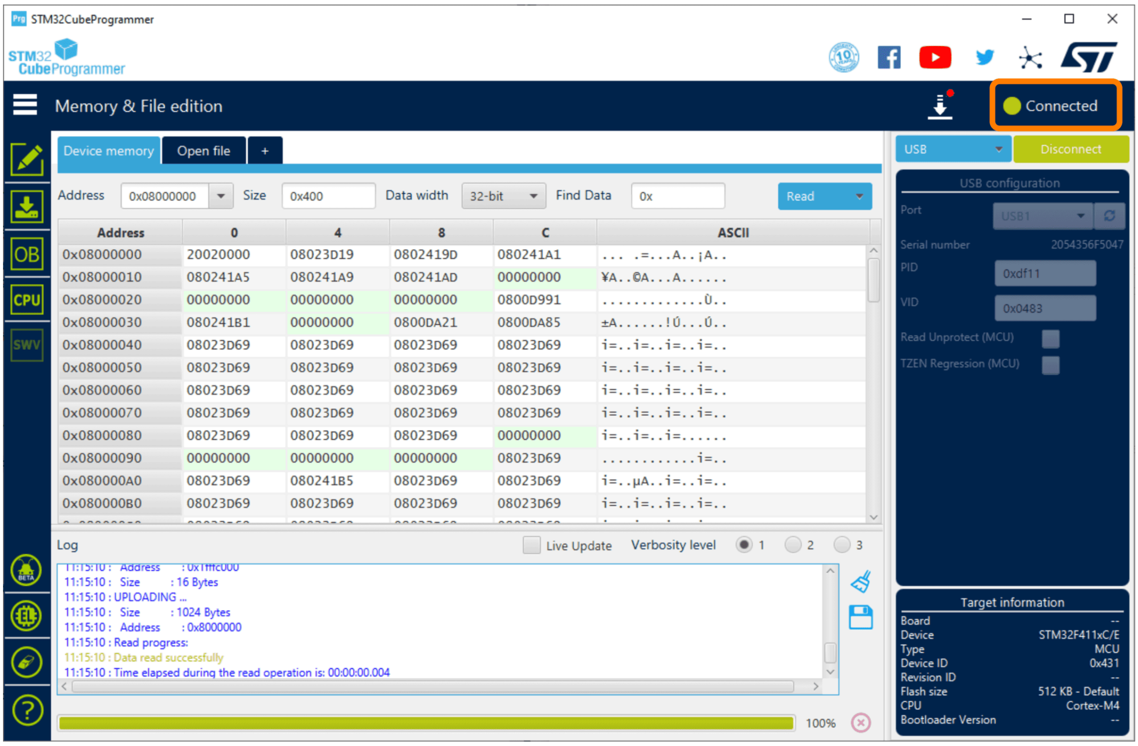


(a) Open the dropdown menu and select USB. The panel shown in the next image opens on the right.

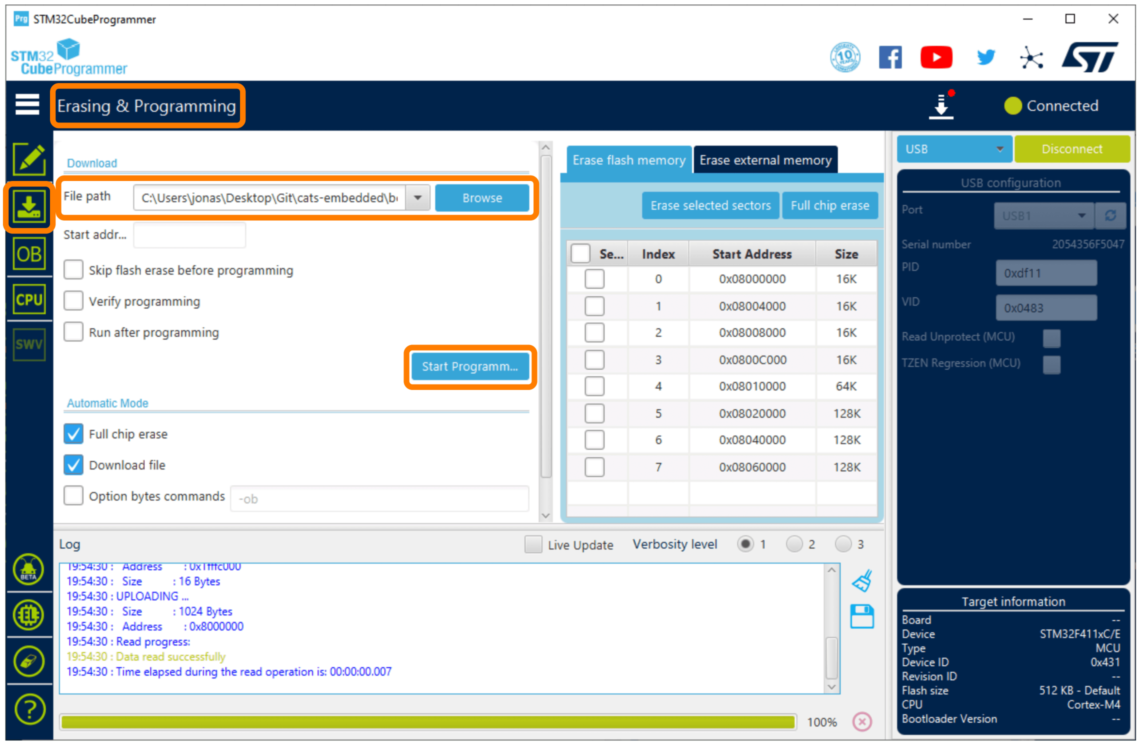


(b) Select the displayed USB port in the Port field, then select Connect.

Figure 12: Flashing new software to the board.

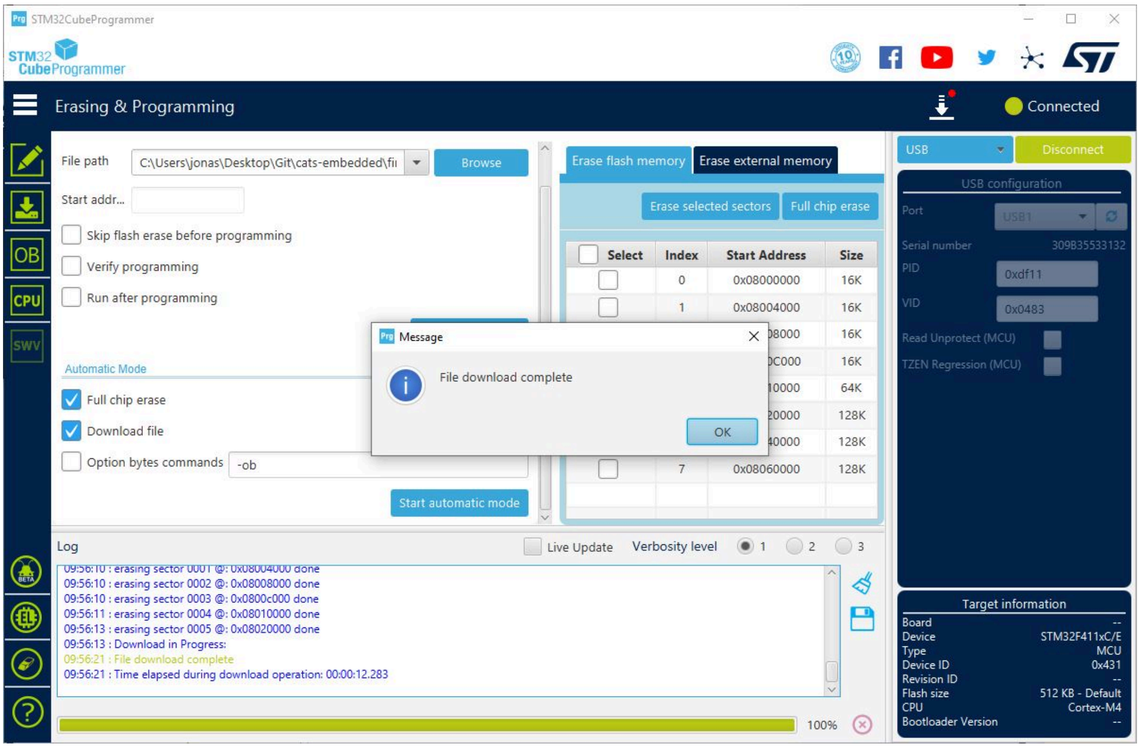


(c) Confirm that the Connected status appears in the upper-right corner.



(d) Select Erasing & Programming on the left. Use Browse to select the appropriate `.bin` file, then select Start Programming.

Figure 12: Flashing new software to the board (cont.).



(e) This message appears when the firmware has been flashed successfully.

Figure 12: Flashing new software to the board (cont.).

5.4 Beeping Patterns

The CATS Vega flight computer uses beeping patterns to indicate its current state or a potential error. The tables below list the available patterns.

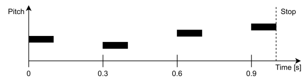
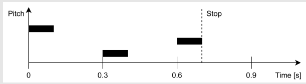
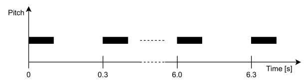
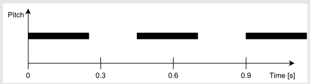
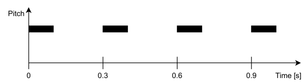
State	Description	Pattern
Bootup	The flight computer has booted up.	
Calibrating → Ready	The flight computer has switched from the Calibrating state to the Ready state.	
Ready	Calibration was successful, and the flight computer is in the Ready state.	
Testing	The flight computer is in testing mode. This pattern sounds only after the computer is rebooted.	
Testing Armed	The flight computer is in the armed testing state.	

Table 8: Overview of state beeping patterns.

Note: The pitch axis shows only relative, unitless changes.

Error	Description	Pattern
Filter Error	A Barometer or accelerometer error is present. If this error persists, **do not fly; the flight computer's functionality is compromised.**	TBD
Pyro Error	A configured pyrotechnic charge is not detected. If this error persists, **do not fly; the configured recovery mechanism will not work.**	TBD
Log Full	The flash chip is full. If you fly, the flight will not be recorded.	TBD
Telemetry Hot	The telemetry chip has reached 60 °C and may be damaged.	TBD
Calibration Error	The calibration is faulty. **Do not fly!** Return to the rocket and reboot the flight computer to restart calibration.	TBD

Table 9: Overview of error beeping patterns.

Note: Bad calibrations usually happen when the flight computer is turned on and then rotated. Only turn the flight computer on once the rocket is in launch configuration and upright on the launch pad.

Note: Audible error codes are not currently implemented. Errors are shown only through telemetry on the Ground Station. Audible error codes will be added in a future update.

6 Ground Station

The Ground Station is the counterpart to the CATS Vega. It receives data from the flight computer and sends commands to it. It displays the rocket's position, velocity, system health, and other important information in real time. This chapter explains how to use the Ground Station and describes its operating principle.

Figure 13: Ground Station

6.1 Hardware

6.1.1 Overview

The Ground Station is built around an ESP32-S2 microcontroller and features a transflective display that remains readable in bright sunlight. The onboard flash can store up to 1 MB of data, enough to track over an hour of flight data.

6.1.2 Specifications

Microcontroller	ESP32-S2
Flash Memory	1 MB
Battery	Li-Ion 18650
Power Consumption	60mA
Charging Current	500mA
Screen	LS027B7DH01
Radio	2x SX1280
Radio Range	Tested to 10km @100mW
GNSS	ATGM336H-5N

Table 10: Ground Station Specifications

6.2 How to Use

This section covers the basic use of the Ground Station. For more advanced information, refer to the later sections.

6.2.1 Explanation of All Menus

Use the joystick to move left, right, up, and down. Press the A button to open a menu or select an option, and press the B button to go back.

Live Data

The Live Data screen shows all data received from the Vega flight computer, together with information about the data-link quality.

Figure 14: Live Data of the Ground Station.

The current flight-computer status appears at the top of the screen. The rocket's altitude, vertical velocity, GNSS coordinates, battery voltage, pyro continuity, and errors appear below it.

At the bottom, information about the telemetry link is displayed:

- **AGE** - The packet age in seconds. If no packet is received for 5 seconds, the link disconnects.
- **SNR** - Signal-to-noise ratio in dB. The link can remain active down to an SNR of −15 dB. A low SNR indicates substantial radio interference at your location.
- **LQ** - Link quality as a percentage: the ratio of packets received to packets expected during the last 3 seconds.
- **RSSI** - Received signal-strength indication in dBm. The link can remain active down to an RSSI of −110 dBm. As a rule of thumb, RSSI decreases by 6 dB each time the distance doubles.

Recovery

The Recovery screen helps you locate the rocket after it lands. It uses the rocket's last known GNSS location, or its current location if the connection is still active. The Ground Station's onboard sensors calculate the distance and direction to the rocket. For accurate direction guidance, calibrate the device's compass outdoors near the launch site and away from large metal objects. The calibration procedure will be explained in a later version of this manual.
The Recovery screen shows the GNSS coordinates of the Ground Station and the rocket, as well as the distance to the rocket. Follow the arrow to locate the rocket.

Testing

The Testing screen is used to perform manual tests. Refer to Section 8 for a detailed explanation of testing mode and how to use it.

Data

Not yet implemented.

Sensors

The Sensors screen displays raw data from the onboard IMU, magnetometer, and GNSS module. It also allows you to calibrate the magnetometer.
Calibrate the magnetometer if the compass on the Recovery screen does not point north. On the Sensors screen, press A and follow the instructions. Slowly rotate the Ground Station in every direction while the progress appears on the screen. When progress reaches 100%, the calibration is complete and is saved on the Ground Station.

Settings

The Ground Station settings are mostly self-explanatory, and tooltips are provided for every option. The current software version supports the following settings:

- **Timezone**: Choose your local timezone to display the correct time.
- **Stop Logging**: Choose whether the Ground Station stops logging after touchdown or continues logging indefinitely.
- **Version**: Ground Station software version number.
- **Bootloader**: Start the bootloader for software updates, as described in Section 6.2.6 .
- **Telemetry Mode**: Select Single or Dual mode, as described in Section 6.2.2 .
- **Link Phrase 1**: Set the phrase that must match the link phrase configured on the CATS Vega.
- **Link Phrase 2**: Set the phrase that must match the second CATS Vega when tracking two flight computers. This setting is not used in Single mode.
- **Testing Phrase**: Set the phrase that must match the testing phrase configured on the CATS Vega. This phrase is required to use the testing mode described in Section 8 .

6.2.2 Telemetry Modes

The Ground Station's telemetry settings include a *mode* option. Because the Ground Station has two receivers, it supports two modes.

In **Dual mode**, the Ground Station can track two Vega flight computers. This is useful when a section separates from the rocket and you want to track it as well as the main body.

In **Single mode**, the Ground Station tracks one Vega flight computer. Packets from both receivers are combined, allowing more data to be received than with a single receiver. For best diversity performance, use one directional antenna and one omnidirectional antenna.

6.2.3 Data Streaming via USB

When connected to a computer via USB, the Ground Station continuously streams each newly received telemetry packet through its virtual serial port. Each line identifies the radio link and includes the timestamp, flight state, GPS coordinates, altitude, velocity, and battery voltage; in dual-receiver mode, data from both links is reported.

The serial stream emits one line per newly received telemetry packet. All units are fixed and are not affected by the Ground Station's unit settings:

- **Link**: Receiving radio link number (1 or 2)
- **Ts**: Flight-computer uptime in seconds, with 0.1 s resolution

- **State**: Numeric flight state (0–7: Invalid, Calibrating, Ready, Thrusting, Coasting, Drogue, Main, Touchdown)
- **Lat** / **Lon**: GPS coordinates in decimal degrees
- **Alt**: Estimated altitude in meters
- **Vel**: Estimated vertical velocity in meters per second
- **V**: Flight-computer battery voltage in volts, with 0.1 V resolution

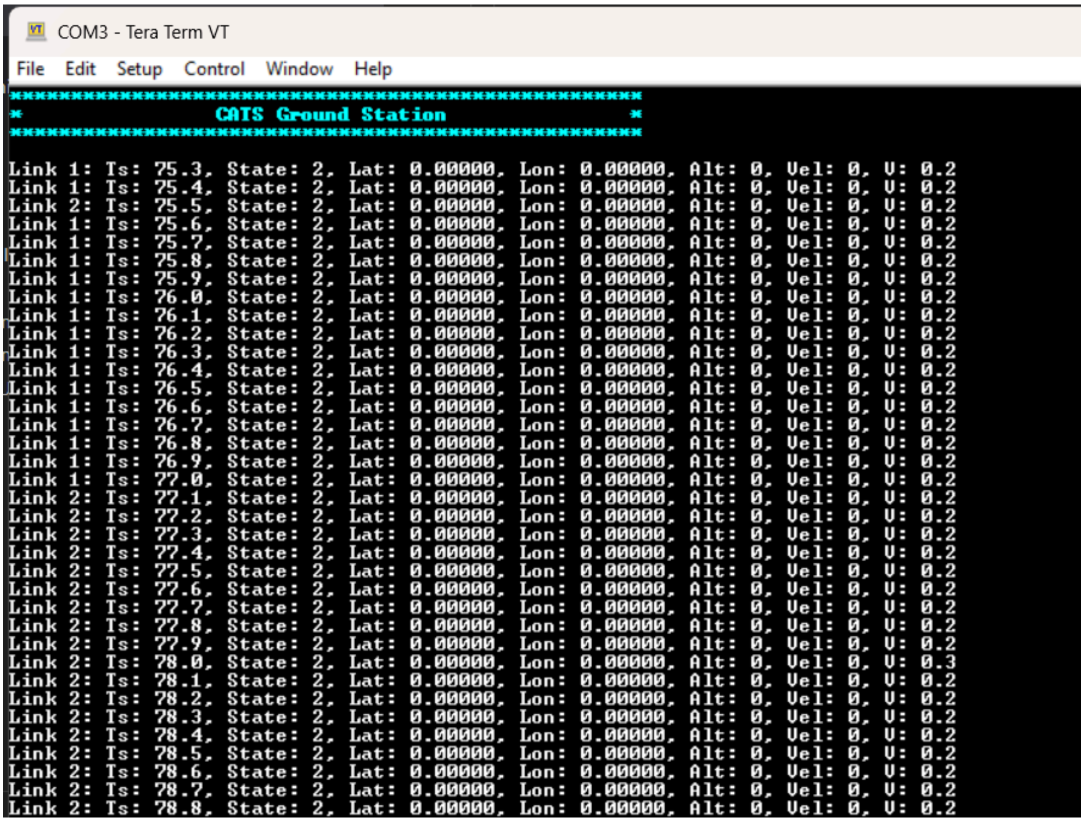


Figure 15: Ground Station telemetry data from both radio links streamed over the USB serial port.

6.2.4 Charging

The Ground Station is powered by a Li-ion 18650 battery. A fully charged battery provides more than 8 hours of operation. Charge the battery through the USB port. At a charging current of 500 mA, a full charge can take up to 6 hours. The LED next to the USB port lights while the battery is charging and turns off when charging is complete. To replace the internal battery, remove the battery cover on the back of the Ground Station. If the replacement battery has different specifications, the estimated remaining charge shown on the screen may differ from the actual percentage.

6.2.5 How to Get the Data on Your Computer

Like the CATS Vega, the Ground Station is recognized as a mass-storage device when connected to a computer. Open the device folder and drag the recorded logs to your preferred location. Ground Station logs are stored as `.csv` files.

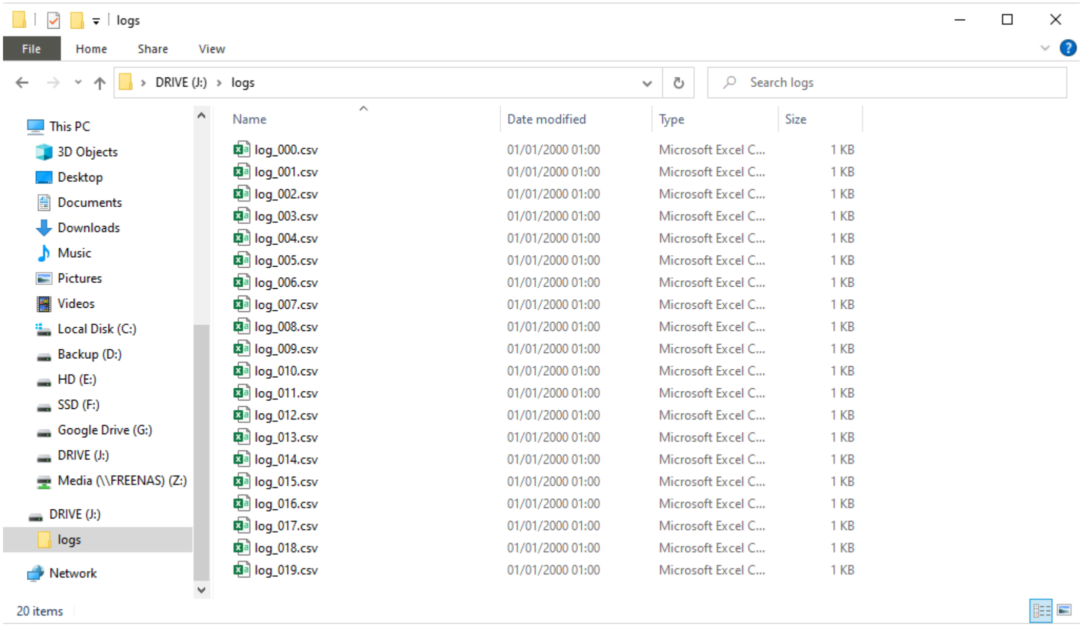


Figure 16: Ground station data when connecting the ground station to the user computer.

6.2.6 Software Updates

To update the Ground Station software, enter DFU mode. First, connect the Ground Station to your computer. In the Settings panel, select Bootloader. A large USB symbol appears on the screen, and a mass-storage device named SAOLA1RBOOT appears on the computer. Drag the firmware file into this folder to overwrite the old file.
You can also enter the bootloader using the hardware controls. Remove the Ground Station casing and connect the Ground Station to your computer. Quickly press the reset button, followed by the boot button. The SAOLA1RBOOT mass-storage device will appear on your computer. Drag the firmware file onto the device.
Firmware filenames end in `.UF2`. The latest Ground Station firmware can be downloaded from our repository.

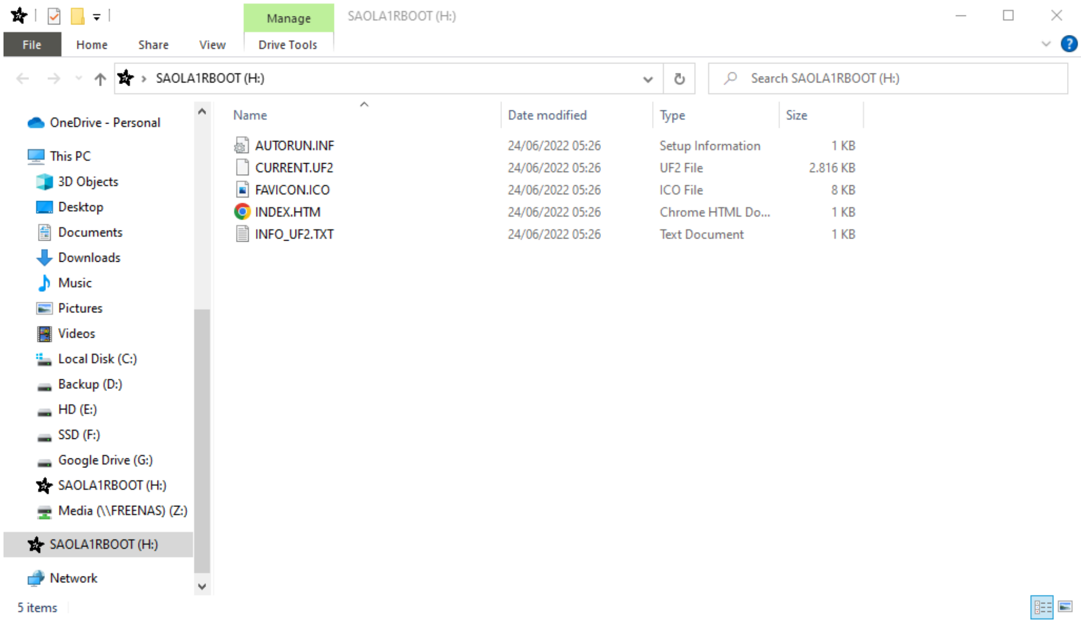


Figure 17: SAOLA1RBOOT mass storage device when successfully changing to the DFU Mode

7 Example Configuration of the CATS Vega + Ground Station

This chapter presents two example configurations, showing both the relevant Configurator settings and the corresponding hardware setup.
The first example is a simple configuration with a drogue chute and a main chute. The second example is intended for more advanced rockets and includes custom events.

7.1 Simple Example Configuration

For the simple example, assume the following configuration:

1. The drogue chute is deployed by a pyrotechnic charge connected to Pyro Channel 1, which must be turned on (Fig. 19b).
2. The expected time to Apogee is 15 seconds and is used for a backup timer (Fig. 19c).
3. The main chute is deployed by a pyrotechnic charge connected to Pyro Channel 2, which must be turned on (Fig. 19b). No timer is configured for main-chute deployment.
4. The liftoff acceleration is simulated to be 70 $\frac{m}{s^2}$ (Fig. 19a).

Configure these settings on the flight computer, then connect the battery, switch, and pyrotechnic charges to Pyro Channels 1 and 2. The completed CATS Vega setup should match Fig. 18 .

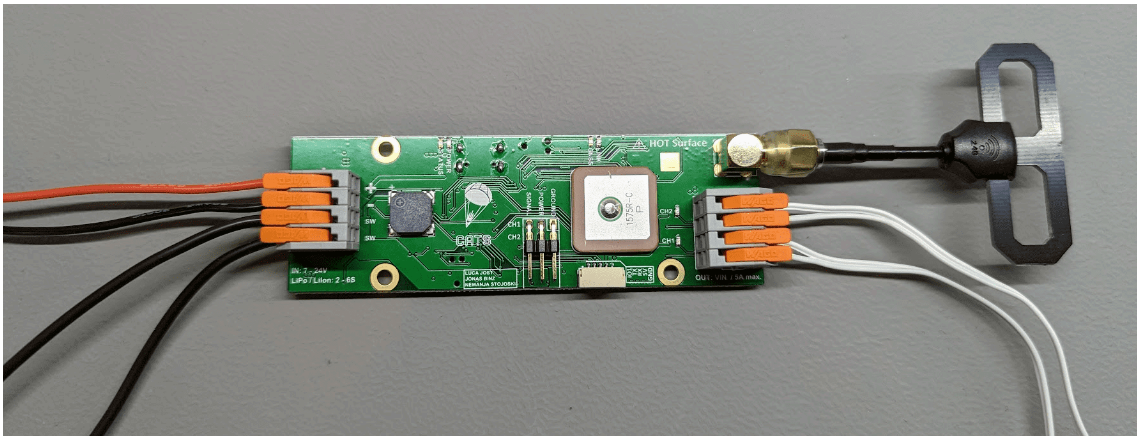
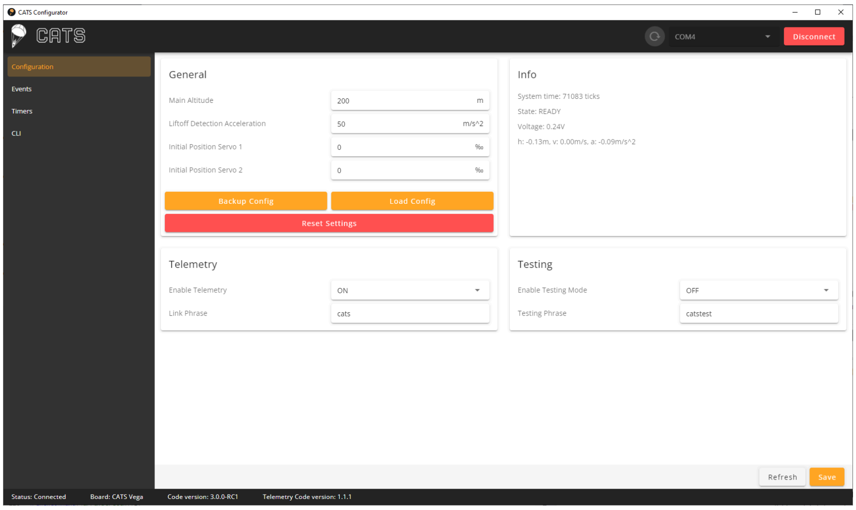
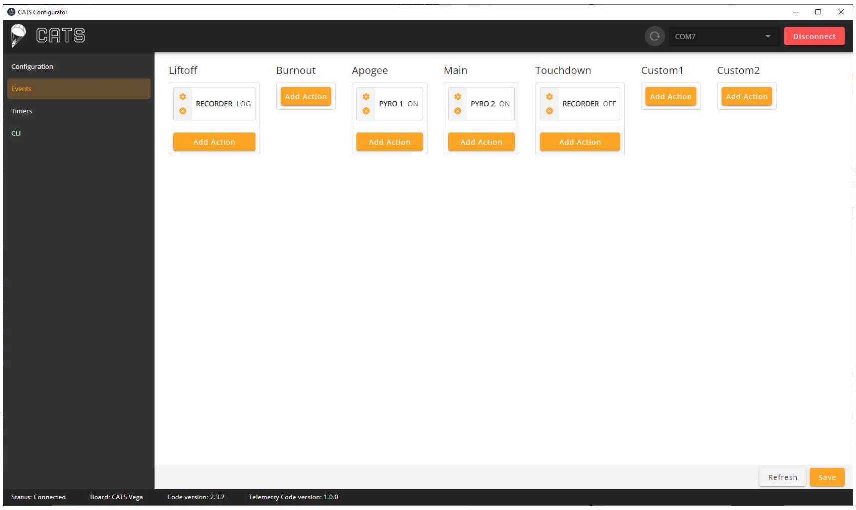


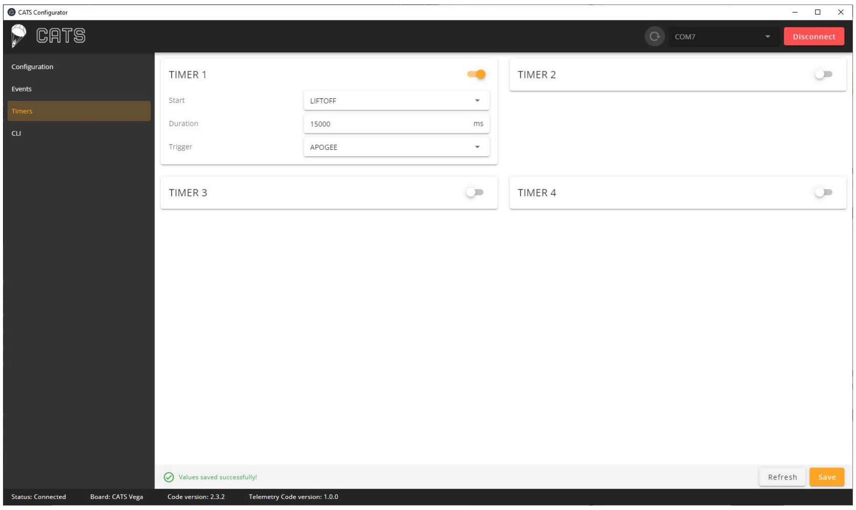
Figure 18: CATS Vega with a battery, switch, Pyro 1 and Pyro 2 connected.



(a) Configuration Tab (simple example).



(b) Event Tab (simple example).



(c) Timer Tab (simple example).

Figure 19: Configurator screenshots for the simple example configuration.

7.2 Advanced Example Configuration

For the advanced example, assume the following configuration:

1. The drogue chute is deployed by a solenoid valve connected to Pyro Channel 1. The channel must be turned on for two seconds and then turned off (Fig. 21b).

2. The expected time to Apogee is 45 seconds and is used for a backup timer (Fig. 21c).

3. The main parachute is deployed by Servo Channel 1. Its initial position is 15 degrees, and the deployment position is 90 degrees (Fig. 21a , Fig. 21b).

4. No main parachute backup timer is used (Fig. 21c).

5. The main parachute should be opened at 350 m (Fig. 21a).

6. The liftoff acceleration is simulated to be 60 $\frac{m}{s^2}$ (Fig. 21a).

7. A camera is connected to Pyro Channel 2. It turns on at liftoff and off at touchdown (Fig. 21b).

8. The low-level I/O is turned on at burnout (Fig. 21b).

Configure these settings on the flight computer, then connect the battery, switch, solenoid valve to Pyro Channel 1, camera to Pyro Channel 2, and Servo to Servo Channel 1. The completed CATS Vega setup should match Fig. 20 .

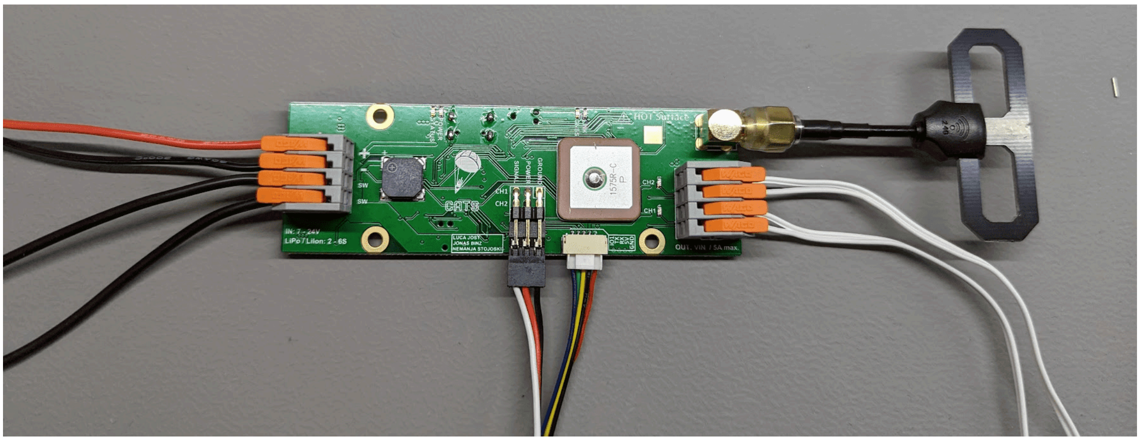
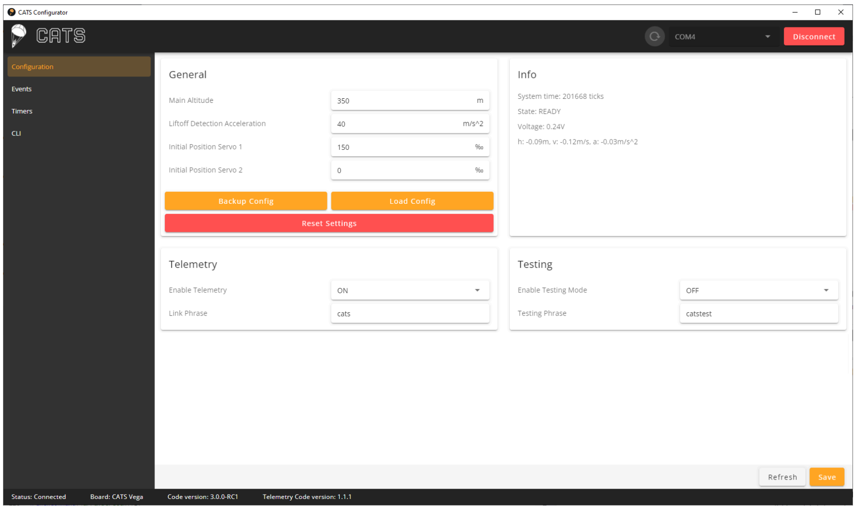
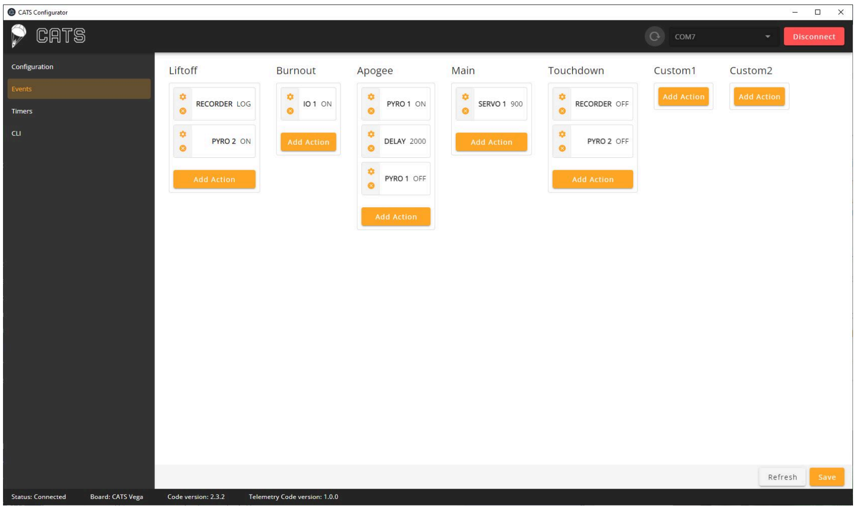


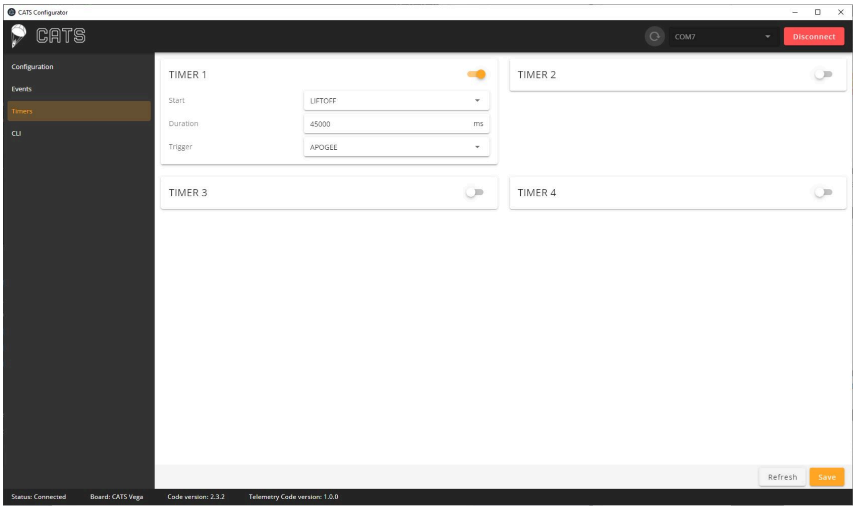
Figure 20: CATS Vega with a battery, switch, solenoid valve, camera and servo connected.



(a) Configuration Tab (advanced example).



(b) Event Tab (advanced example).



(c) Timer Tab (advanced example).

Figure 21: Configurator screenshots for the advanced example configuration.

8 Testing

This section explains how to run tests with the CATS System.

8.1 Working Principle

The testing mode of the Vega flight computer works only with a ground station. Once the flight computer is in this mode all processing of information is disabled. This means that if the flight computer is flown while in testing mode, make sure you bring a shovel with you since you will need it to recover your rocket.
Once the flight computer is in testing mode, a beeping sound is emitted, signaling that the computer is indeed in testing. A command from the ground station needs then to be set to arm the flight computer. When the flight computer is armed, events can be executed from the ground station.

> **Note:** Unlike during flight, events can be triggered several times without needing to reboot the system.

By design, the testing mode only enables triggering events and not actions so that the user can verify that they added the correct configuration to the flight computer.

8.2 Enabling the Testing Mode

To enable the testing mode, follow these steps:

1. Connect your CATS Vega to your computer, start the Configurator, and connect to the flight computer.
2. On the Configurator's Home screen, enable Testing Mode.
3. Set a testing phrase.
4. Reboot the flight computer. If the flight computer is not restarted, the testing mode is not activated.
5. The flight computer should emit the "Testing" beeping pattern. See Section 5.4 for more information.
6. Turn on your Ground Station.
7. Make sure that the link phrase and testing phrase match those configured on the flight computer.
8. Open the Testing menu on the Ground Station.

Figure 22: Ground Station Main Menu.

9. Read the disclaimer carefully and arm testing mode. **Attention:** After this step, executing events will trigger the connected mechanisms. Follow all safety guidelines.

Figure 23: Arming the test mode.

10. A pop-up indicates that testing mode is being activated. Wait until it disappears.

Figure 24: Waiting for testing mode to activate.

11. The flight computer should emit the "Armed Testing" beeping pattern. See Section 5.4 for more information.
12. Select the event that you want to trigger.

Figure 25: Selection of the Event to be triggered.

13. Select the event and confirm.

Figure 26: Confirming to trigger the desired Event.

Warning: Testing mode enables manual triggering of events and corresponding actions associated with them. This feature should only be used for testing purposes and never during flight or other non-controlled activities. CATS GmbH is not responsible for any potential injuries or material damage caused by manual operation of the CATS System.

9 Advanced Information

This section explains the core components of the CATS Vega board in greater detail. It is intended for advanced users.

9.1 Software Overview

This section provides a brief overview of the software architecture. It is intended for advanced users with some programming experience; understanding it is not required to use the flight computer. The software is implemented in C++ and uses FreeRTOS as its foundation. The hardware is initialized first, after which the tasks are started. Figure 27 shows the running tasks.

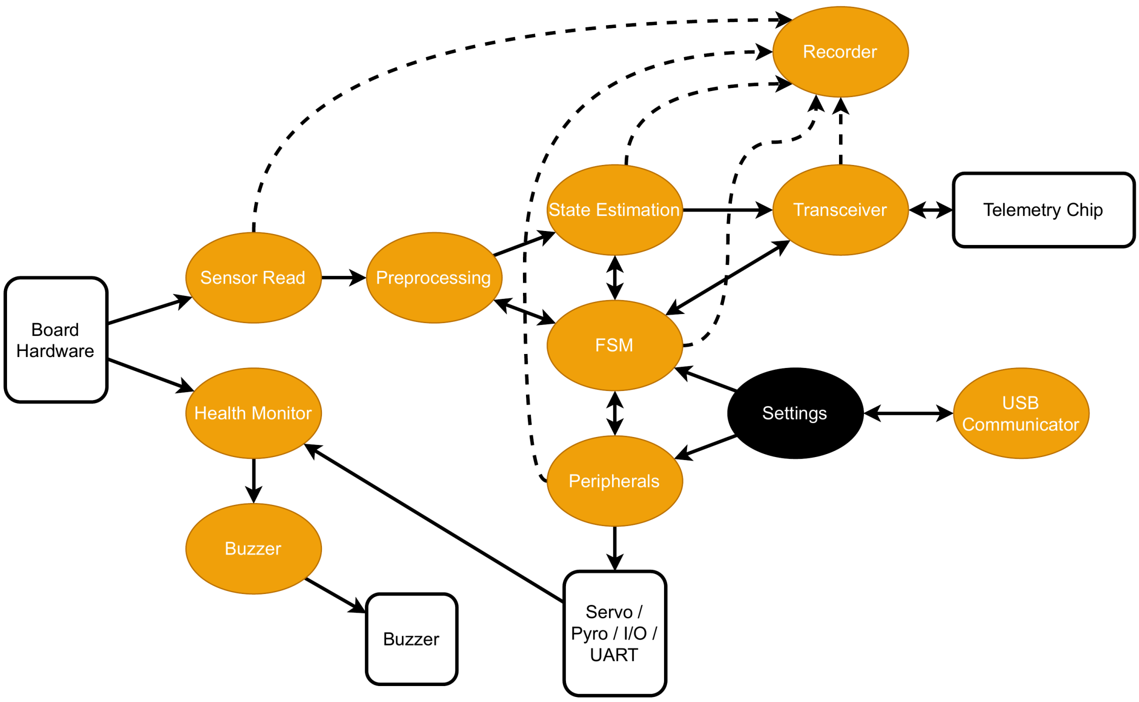


Figure 27: Illustration of the different FreeRTOS tasks interacting with each other and the hardware. The black circle 'Settings' is just a memory region that is being accessed by different tasks.

The following list briefly describes each task.

- **Sensor Read** reads the Barometer and IMU data and provides it to the Preprocessing task.
- **Preprocessing** converts the raw barometric pressure to altitude above ground level and the linear acceleration to acceleration in the "up" direction. It provides the processed data to the FSM and State Estimation tasks.
- **State Estimation** estimates the current height and velocity based on the Preprocessing task's output and the FSM state.
- **FSM** computes the current flight phase from the State Estimation and Preprocessing outputs.
- **Transceiver** handles communication between the main chip and the telemetry chip.
- **USB Communicator** is a group of three tasks that handle the interface between the flight computer and the user's computer. These tasks do not start when USB is not connected.
- **Peripherals** triggers all user-defined events.
- **Health Monitor** checks values like the buzzer and the battery voltage.
- **Recorder** uses a queue to record all data to the flash chip.
- **Buzzer** actuates the buzzer.

9.2 Telemetry

The telemetry system uses 2.4 GHz LoRa and FHSS (Frequency-Hopping Spread Spectrum). FHSS makes transmissions more resistant to interference and more difficult to intercept. It also allows more devices to use the same frequency band with little or no effect on link quality.

Hopping Pattern

The link phrase defines the hopping pattern. It is hashed with a CRC−32 algorithm, and the resulting value seeds a pseudo-random number generator. The generator runs 20 times to define the hopping pattern. As a result, a given link phrase always produces the same pattern. The transmitter and receiver must use the same link phrase to communicate.

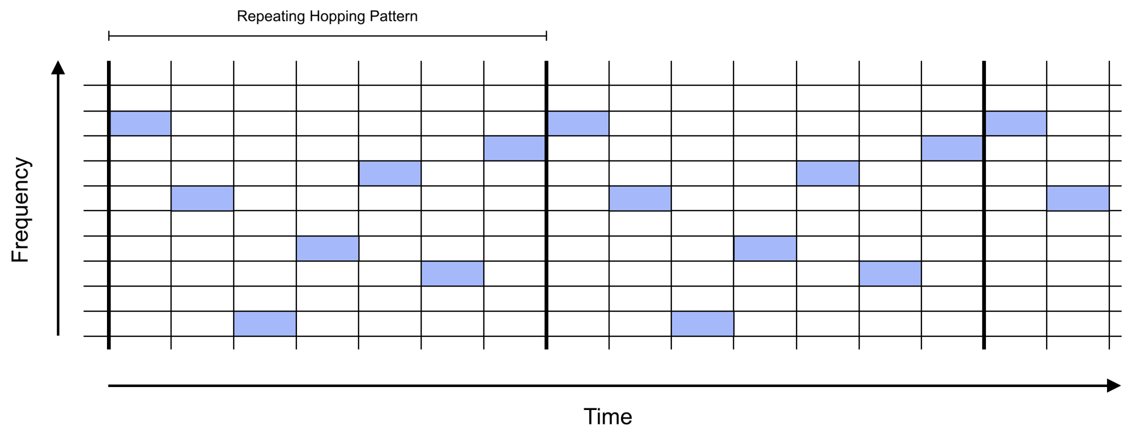


Figure 28: FHSS transmission example

Synchronization

The receiver waits on the first frequency until it receives a synchronization packet. This packet contains the link CRC, which identifies the transmission source. If the remote CRC matches the local value, the receiver hops to the next frequency and waits for data. Each data packet contains a checksum for validating its contents. The receiver measures the interval between packets and hops to the next frequency when a packet is not received within the estimated interval. It can perform 30 hops without receiving a packet before synchronization is lost. If the connection is lost, the receiver returns to the first frequency.

9.3 Estimation Algorithms

State estimation calculates the rocket's velocity and altitude from barometric pressure and linear acceleration in the z direction.

Calibration of Sensors

Linear acceleration is calibrated when the system enters the Ready state. This allows the flight computer to be mounted in any orientation. The gravity vector is used to calculate the up direction, which is then used throughout the flight.

> **Warning:** Power up the flight computer only after the rocket is upright on the launch pad. To prevent repeated transitions into and out of the Ready state, calibration is performed only once, as soon as no motion is detected after startup.

During Calibrating and Ready, the current altitude above sea level is continuously estimated. Altitude above ground level, the value used during flight, is calculated from the altitude above sea level. This calculation assumes that barometric pressure changes very slowly. When Liftoff is detected, the altitude above sea level is fixed, and only the altitude above ground level is updated. **Kalman Filter**

A Kalman Filter estimates altitude and velocity from the calibrated values. Its derivation is described below. We define the state and noise as

$$x(t) = \begin{pmatrix} h(t) \\ v(t) \\ a_o(t) \end{pmatrix} \quad v = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$$

where h is the altitude above ground level, v is the vertical velocity, and a_0 is the estimated offset of the measured vertical acceleration. v_1 is the noise applied to the vertical acceleration, and v_2 is the noise used to estimate the linear-acceleration offset.
The system input is $u(t)$, the linear acceleration measured in the z direction.

$$\dot{x}(t) = \begin{pmatrix} v(t) \\ a(t) \\ \dot{a_0}(t) \end{pmatrix} = A \cdot x(t) + B \cdot u(t) + G \cdot v(t) = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \cdot \begin{pmatrix} h(t) \\ v(t) \\ a_o(t) \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \cdot u(t) + \begin{pmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} v_1(t) \\ v_2(t) \end{pmatrix}$$

This is discretized using a first-order approximation:

$$A_d = e^{AT_s} \quad B_d = \int_0^{Ts} e^{A_d t} \delta t \cdot B \quad G_d = \int_0^{Ts} e^{A_d t} \delta t \cdot G$$

which gives the system

$$x(k+1) = A_d \cdot x(k) + B_d \cdot u(k) + G_d \cdot v(k)$$

The process measurement noise matrix becomes

$$Q(k) = \begin{pmatrix} Q_{\text{acc}}(k) & 0 \\ 0 & Q_{\text{acc\_0}}(k) \end{pmatrix}$$

The measurement step assumes that altitude has already been calculated from barometric pressure using the standard barometric formula.

$$z(k) = h_{\text{meas}}(k) + \omega(k) = \left(\frac{p(k)}{p_0}^{\frac{1}{5.257}} - 1 \right) \cdot \frac{T_0 + 273.15}{L} - h_0 + \omega(k)$$

where $L = -0.0065$, $T_0 = 15$ C, $p_0 = 101250$ Pa, h_0 is the calibrated altitude above sea level, and ω is the measurement noise.

The measurement function is

$$z(k) = H \cdot x(k) = (1 \;\; 0 \;\; 0) \cdot \begin{pmatrix} h(k) \\ v(k) \\ a_o(k) \end{pmatrix}$$

The measurement-noise matrix becomes a scalar:

$$R(k) = R_{\text{height}}$$

The standard Kalman-filter equations can then propagate the state.

Gain Scheduling

Gain scheduling reduces reliance on the barometer during high-velocity flight, because barometric measurements can behave unpredictably in the transonic regime.
At Liftoff and while the rocket is moving quickly, the accelerometer is weighted more heavily when estimating altitude and velocity. At lower velocities, barometric pressure is weighted more heavily because accelerometer drift affects the estimate. As the rocket arcs over, the quality of the accelerometer measurement also decreases.
Two variables control the relative trust in the sensors: Q_{acc} and R_{height}. In the algorithm, Q_{acc} remains constant, while R_{height} changes during flight.
The conditions for changing R_{height} are shown below.

$$R_{\text{height}} = \begin{cases} R_{\text{initial}} & \text{for state} = \text{MOVING or IDLE} \\ R_{\text{max}} & \text{for state} = \text{LIFTOFF} \\ R_{\text{max}} \cdot f(v) & \text{for state} = \text{COASTING} \\ R_{\text{initial}} & \text{otherwise} \end{cases}$$

where $f(v)$ depends on the current velocity.
This gain scheduling effectively filters unexpected barometric measurements at high velocities.

9.4 Using the Python Plotting Tool

For more control over flight data, use the legacy Python plotting tool. It provides the Configurator's plotting functionality in a form that is easier to customize.
To visualize logs recorded by the Vega flight computer, clone the cats-logs repository and run the log_visualizer.py script. Git and Python 3 must be installed to download and run the script.

To visualize your logs, follow these steps:

```
git clone https://github.com/catsystems/cats-logs.git
cd cats-logs/log_parsing
pip install -r requirements.txt
python log_visualizer.py -i <path to input log> -o <path to output directory>
```

The script parses the log file and generates a self-contained HTML file containing all plots. It also generates raw and processed CSV files for each type of recorded value.

For more information about the visualizer script, run:

```
./log_visualizer.py --help
```

9.5 Description of the CLI

This section describes all commands available in the CLI. To access the CLI, connect the board to your computer, connect through the Configurator, and open the CLI tab.
In the list below, square brackets `[ ]` indicate an optional argument, while angle brackets < > identify a parameter name.
After changing the configuration through the CLI, verify it with the `config` command.

`bl`	Put the board into DFU mode	Needed for software updates (refer to Section 5.3.8)
`cd`	Change the current working directory	
`config`	Print the flight config in a human-readable format	
`defaults [–no-outputs]`	Reset to default settings	The default configuration triggers pyro channels. If `–no-outputs` is passed, pyro triggering is not configured.
`dump`	Print configurable settings in a readable format	
`flash_erase`	Erase everything on the flash chip; this might take a while	
`flash_test`	Test writing to and reading from the flash	For testing purposes only; do not use
`flash_start_write`	Start writing to flash	For testing purposes only; do not use
`flash_stop_write`	Stop writing to flash	For testing purposes only; do not use
`flight_dump < flight_id>`	Print a specific flight in binary format	
`flight_parse < flight_id>`	Print a specific flight in a human-readable format	
`get [< variable>]`	Get a variable value, described in Table 12	
`help [< command name>]`	Display all commands with a description	
`lfs_format`	Reformat the flash filesystem	
`log_enable`	Enable log output on the terminal	
`ls [< path>]`	List all files in the current working directory	
`reboot`	Reboot the flight computer	This command does not save the changed settings by itself
`rec_info`	Get information about flash usage	
`rm [< path>]`	Remove a file	
`save`	Save flight configuration	
`set [< variable> =< value>]`	Set a variable, described in Table 12	

stats < flight_id>	Print flight statistics	
status	Show current sensor data, flight phase, and other important information	
version	Show the firmware version	

Table 11: Exhaustive List of CLI Commands

9.5.1 Get and Set Commands

The variables below can be read with the **get** command or changed with the **set** command. Changes are saved to the flight computer's configuration only after the **save** command is run.

`acc_threshold`	Acceleration threshold above which liftoff is detected	
`main_altitude`	Altitude above ground level under which main deployment is triggered	
`timer1_start`	Event which triggers timer 1	
`timer1_trigger`	Event which is triggered once timer 1 elapses	
`timer1_duration`	Timer 1 duration	
`timer2_start`	Event which triggers timer 2	
`timer2_trigger`	Event which is triggered once timer 2 elapses	
`timer2_duration`	Timer 2 duration	
`timer3_start`	Event which triggers timer 3	
`timer3_trigger`	Event which is triggered once timer 3 elapses	
`timer3_duration`	Timer 3 duration	
`timer4_start`	Event which triggers timer 4	
`timer4_trigger`	Event which is triggered once timer 4 elapses	
`timer4_duration`	Timer 4 duration	
`ev_calibrate`	Set the actions associated with the calibration event	Do not use!
`ev_ready`	Set the actions associated with the ready event	Do not use!
`ev_burnout`	Set the actions associated with the burnout event	Do not use!
`ev_apogee`	Set the actions associated with the apogee event	Do not use!
`ev_main_deployment`	Set the actions associated with the main deployment event	Do not use!
`ev_touchdown`	Set the actions associated with the touchdown event	Do not use!
`ev_custom1`	Set the actions associated with the custom 1 event	Do not use!
`ev_custom2`	Set the actions associated with the custom 2 event	Do not use!
`servo1_init_pos`	Set the initial position of servo 1	
`servo2_init_pos`	Set the initial position of servo 2	
`tele_link_phrase`	Set the telemetry link phrase	
`tele_test_phrase`	Set the testing phrase	
`tele_power_level`	Set the telemetry power level	

`tele_adaptive_power`	Enable or disable adaptive power for the telemetry power level	Adaptive power mode boosts output power to maximum when the flight computer is in *THRUSTING* mode and returns it to the user-set value when *TOUCHDOWN* is registered.
`buzzer_volume`	Set the buzzer volume	
`battery_type`	Set the battery type used with the CATS Vega	
`rec_elements`	Set the desired recorded elements	A bit mask corresponding to the `rec_entry_type_e` enum
`rec_speed`	Set the desired sampling period for recording	

Table 12: Exhaustive List of parameters used in the `get` and `set` Commands

10 FAQ

My testing mode doesn't work

To enable testing mode, make sure that you have enabled testing mode on the flight computer through the Configurator and set a valid testing phrase. Otherwise, the flight computer will start in its default mode and will not accept testing commands from the Ground Station.

The Configurator sometimes doesn't apply my settings

Connect the flight computer directly to your computer without a USB hub. If that does not help, contact us on Discord or submit an issue on GitHub.

To be extended with user feedback…